



THE IMPACT OF THE CHURCHILL RIVER DIVERSION ON LAKE STURGEON IN THE LOWER CHURCHILL RIVER

PREPARED FOR:
TATASKWEYAK CREE NATION



Executive Summary

Tataskweyak Cree Nation has long maintained that the Churchill River Diversion (CRD) has profoundly disrupted their traditional way of life by severely degrading the ecological integrity of the lower Churchill River. Since the CRD's implementation, Elders and resource users have documented the collapse of the river's fishery and broader ecosystem. The disappearance of Lake Sturgeon is regarded as a sentinel event, a "canary in the coal mine", signaling broader ecological harm that extends to other species and ultimately to human well-being. The loss of habitat for fish, migratory birds, and riparian wildlife has significantly curtailed fishing, hunting, trapping, gathering, and recreational activities along the river. Today, Lake Sturgeon persist in only one pre-CRD habitat, while other species such as brook trout, Arctic grayling, beaver, muskrat, and riparian fauna have largely disappeared.

The Committee on the Status of Endangered Wildlife in Canada (COSEWIC) has formally designated Lake Sturgeon in the lower Churchill River as Endangered, with only a single remnant population remaining. This report integrates five complementary lines of inquiry to investigate the causes of this decline:

- Indigenous Knowledge of Lake Sturgeon life history, movement, and habitat use
- Scientific literature on river rehabilitation and environmental flow regimes
- Historical predictions made by scientists prior to CRD implementation
- Empirical data from Lake Sturgeon population, movement, and environmental DNA (eDNA) studies
- Hydraulic modeling of instream flow conditions before and after CRD regulation

Collectively, these investigations demonstrate that the CRD's altered flow regime has driven the collapse of Lake Sturgeon populations through two primary mechanisms:

1. The diversion of over 90% of the river's natural flow for extended durations has drastically reduced the availability of suitable habitat.
2. The introduction of abrupt and irregular flow surges and drops has created a highly unpredictable hydrological environment to which the downstream ecosystem cannot adapt.

Hydraulic modeling reveals substantial differences in wetted cross-sectional area, average depth, and flow velocity between pre- and post-CRD conditions. These findings underscore the urgent need for an environmental flow regime that reconciles the ecological and cultural priorities of Tataskweyak Cree Nation with the operational requirements of Manitoba Hydro.

To support Lake Sturgeon recovery and broader ecosystem restoration, this report recommends the following flow adjustments under the CRD licence:

- Gradual increase in minimum flows from 425 m³/s (15,000 cfs) on May 1 to 708 m³/s (25,000 cfs) by August 1
- Gradual decrease from 708 m³/s (25,000 cfs) on August 1 to 425 m³/s (15,000 cfs) by May 1 of the following year

Additional recommendations include:

- Utilizing the remnant Lake Sturgeon population to reintroduce the species upstream
- Constructing a fish passage facility at Missi Falls
- Establishing a long-term Lake Sturgeon monitoring program led by Tataskweyak Cree Nation
- Conducting five-year reviews to evaluate the effectiveness of recovery measures

Together, these actions constitute a comprehensive strategy to restore Lake Sturgeon and rehabilitate the lower Churchill River ecosystem, while safeguarding the last remaining population of a species that holds deep cultural and ecological significance.

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1. Introduction

Namew – the Cree name for Lake Sturgeon (*Acipenser fluences*) – belong to an order of primitive fish which until quite recently flourished in Canada's largest lakes and rivers for over 200 million years. Among freshwater fish they have a unique life history characterized by a long natural lifespan, growing slowly to over a couple of metres in size, and low reproductive rates including late onset sexual maturity. Diverse habitats – ranging from fast-flowing, well-oxygenated river sections with rocky beds for spawning, to deep slow-flowing waters with finer sediment beds for mature adults – are crucial for the survival and reproduction of Lake Sturgeon. These requirements make them vulnerable to many human influences including loss of habitat, pollution, and overharvesting (CWSFB, 2012; Peterson et al., 2007; Scott and Crossman, 1973). Today, few healthy populations remain in Canada, and many anthropogenic stressors continue to limit the effectiveness of conservation and restoration efforts (Noakes et al., 1999).

In Manitoba, hydroelectric facilities cause the greatest stress for Lake Sturgeon because they impose unnatural flows that alter and destroy aquatic habitat conditions, and they obstruct upstream passage for fish which limits species diversity and access to historic spawning grounds. Almost all of Manitoba's large rivers – the Winnipeg, Saskatchewan, Nelson, Burntwood, and Churchill – are affected by hydroelectric facilities, whose generating stations and control structures alter the natural hydrologic regime and separate and degrade the upstream and downstream ecosystems.

Starting operations in 1976, the Missi Falls Control Structure changed the flow regime of the lower Churchill River by diverting most of its flow into the Nelson River via an excavated channel, called the South Bay Channel, at Southern Indian Lake. This scheme, known as the Churchill River Diversion (CRD) was developed to join Manitoba's two largest rivers – the Churchill and the Nelson – to provide the additional water required to power the five hydroelectric generating stations along the Burntwood and Nelson Rivers.

This study focuses on the operation of the CRD, a major component of hydroelectric power development in Manitoba, and the effect its regulation of the lower Churchill River's flow has on Lake Sturgeon in this section of the river.

The lower Churchill River is 440 kilometres long and flows from Missi Falls at the outlet from Southern Indian Lake to Hudson Bay at the town of Churchill (Fig. 1). The upper 160 km, from Missi Falls to Billard Lake, is composed of a series of lakes joined by short reaches of river. Over this first 160 km, the river drops by about 50 m (170 ft). From the outlet of Billard Lake to Churchill, the river flows for 280 km in a well-defined channel over a series of rapids that drops

180 m (600 ft) to the Bay. Roughly halfway down the river from Missi Falls, a remnant of a once widespread population of Lake Sturgeon survives in a 40 km reach between Mountain and Swallow Rapids (AAE, 2025a and 2025b).

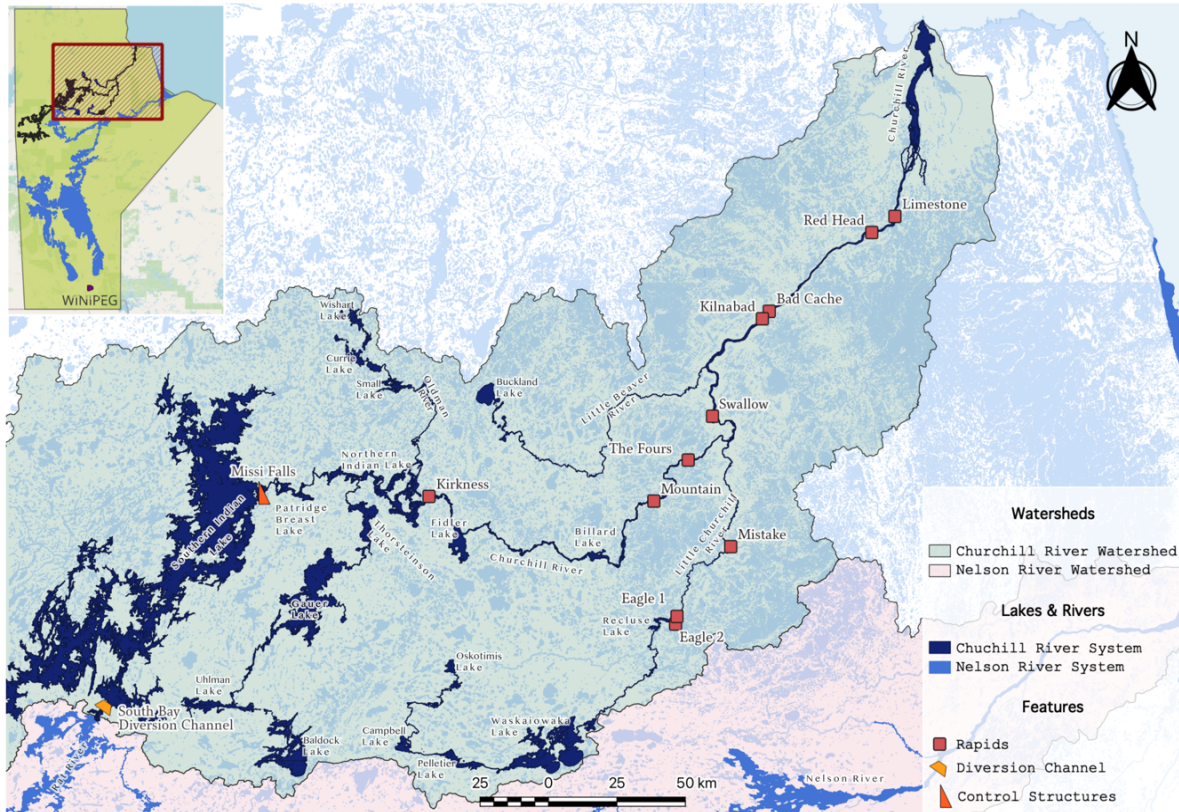


Figure 1: Lower Churchill River.

Hydroelectric power development creates significant challenges to sustainable resource management in Manitoba given its detrimental impacts on fish and the habitats they rely on. It directly conflicts with Fisheries and Oceans Canada's mandate to support the conservation and stewardship of species at risk and to prohibit the death of fish and alteration, disruption or destruction of their habitat. Lake Sturgeon in the lower Churchill River were confirmed as Endangered by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC, 2017). However, little appears to have been done to protect the remaining population or improve habitat conditions for Lake Sturgeon along the river.

The Churchill, Burntwood and Nelson Rivers flow through the traditional lands of the Tataskweyak Cree Nation and are integral to their way of life – but there is little evidence to suggest that provincial and federal regulators give due regard to the traditional knowledge of these people.

1.1. Methodology

To frame properly and analyze the effect of the CRD's operation on Tataskweyak Cree Nation¹, we start in Chapter 2 by exploring Indigenous Knowledge of Lake Sturgeon in the lower Churchill River. Elders, holders of traditional knowledge, and resource users carry generations of experiences and intimate knowledge of their traditional territory's vast lands and waters, long before hydroelectric development began. In addition to valuing and validating this knowledge, our study will address a key focus of the 2019 Fisheries Act expressed in sections 2.4, 2.5, 4.1 (1) and 4.2: to establish strategic and lasting collaboration with Indigenous Peoples. Work of this nature must be understood as part of the broader journey toward Reconciliation. As emphasized by the Truth and Reconciliation Commission of Canada (2015), the first step in this process is to listen—to hear the truths and histories of Indigenous peoples—and to honour that knowledge as foundational. Commissioner Marie Wilson, reflecting on the Commission's work, stated: *"I see you. I hear you. And I believe you,"* underscoring the importance of validating lived experience as a basis for healing and change (University of Alberta, 2016).

Chapter 3 examines how the CRD regulates flow in the lower Churchill River and contrasts the current flow regime with the natural conditions that existed prior to the diversions's implementation. Chapter 4 reviews early scientific assessments that anticipated the potential impacts of the CRD. Chapter 5 presents our research on estimating Lake Sturgeon population size and tracking their movement within the river system. In Chapter 6, we detail the development and results of an instream flow model designed to compare hydraulic conditions before and after the CRD. Finally, Chapter 7 integrates these lines of evidence into a comprehensive discussion, culminating in our conclusions and recommendations for protecting Lake Sturgeon, restoring their habitats, and re-establishing their presence throughout the river.

¹Tataskweyak Cree Nation was formerly known as Spit Lake Cree Nation. The name was officially changed in 1992, but references to Split Lake Cree Nation continue.

2. Indigenous Knowledge

This study respects and incorporates Indigenous Knowledge in determining the cause of the Lake Sturgeon's demise along the lower Churchill River. Over the past nine years, Halket Environmental Consultants Inc. and AAE Tech Services Inc. gathered Indigenous histories and current perspectives of life on the Churchill River prior to and since construction of the CRD. We interviewed current resource users and Elders from Tataskweyak Cree Nation, and reviewed reports by Tataskweyak and other First Nations affected by the CRD. These first-hand accounts described below paint a drastically different picture of the river's former bounty than is the case now. They also provide us with a wealth of information detailing the importance of Lake Sturgeon to First Nation culture and the damage to the Churchill River and Lake Sturgeon post-CRD.

Prior to European contact, the Tataskweyak Cree lived, hunted, trapped, fished and gathered resources on and along the lower Churchill River. Their way of life centered around the lakes and especially the rivers, the "roads" by which they travelled and where wildlife and plants could be harvested. This life was transformed when they were forcibly moved to a central community, Split Lake, in the 1950s. Although now established at Split Lake, Tataskweyak Cree Nation members still hunt, fish, trap, and use the resources throughout their traditional territory. First Nations members interviewed in the 1950s and 1960s described their life before hydroelectric development:

"The testimony of the Elders interviewed for this [Split Lake Cree] study, who remember the life the Split Lake Cree once enjoyed in their tribal homeland, shows that they, more than anyone else, realize what they have lost.

"They speak of a life of freedom and of fulfillment in an environment that they knew and cherished. Many of the Elders recall the time when there was no need for money because the very concept of money was not a part of the Indian way of life. Their life in their resource area was not nomadic but was an accommodation and an arrangement with nature and her rhythms. The people had everything they required – food, shelter, clothing, recreation, medicine, and their relationship with their Creator" (Split Lake Cree First Nation, 1996, p. 18).

Tataskweyak members know their land and waters, and how they have changed since the CRD. They easily point out important Indigenous landmarks on a sparsely marked map of their territory: ancient encampments, burial grounds so old that they are now being reclaimed by the land and waters, and spiritual places. They mark old, favoured Lake Sturgeon spawning grounds and fishing sites, ancestral camps, and current-day water-monitoring stations. Their insights and intimate knowledge of the lakes and rivers on the map come from their experience living off the land and water now and before it was changed:

“More than anything else, the Elders speak of the water as the lifeblood of their existence. Water, before the hydroelectric project, was always clean. People could get water from the lake [Split Lake] and know it was safe. They could fish in the lake near their homes and know that they need not fear the possibility of poison from the fish. Their nets would not be filled with algae and debris. They could land anywhere along the banks of the lakes and rivers; the shorelines had not yet been destroyed and desecrated. They could navigate the lakes and rivers in safety and know what to expect. In the winter time [sic] they would know where to travel, where the ice was safe. These things are no longer possible” (Split Lake Cree First Nation, 1996, p. 18).

Before the CRD, Lake Sturgeon were plentiful and were a traditional country food and a mainstay for Solstice celebrations and family gatherings.

“One of the Elders interviewed described the lower Churchill River prior to the CRD as being beautiful and powerful, and the rivers and lakes in the area as fertile and productive (per. comm., 2012)” (Cree Nation Partners, 2015, p.31).

Lake Sturgeon remain an important component of Cree spiritual life. Cree often refer to large Lake Sturgeon as grandfather, while sacred stories demonstrate both the social and religious value of Cree-Lake Sturgeon relations (Hannibal-Paci, 1998).²

The Lake Sturgeon were also the source of useful raw materials including skin, oil, and isinglass³. Because of the Lake Sturgeon’s significance, Tataskweyak Cree paid close attention to them: their populations, characteristics, seasonal movements, as well as locations of their feeding and spawning grounds.

In *The Story of Sturgeon* by Cree Nation Partners (2015), the commercial Lake Sturgeon fishing industry was examined through historical records and traditional knowledge interviews. The results shed light on their historic abundance and distribution throughout the Churchill River system. The report states that commercial fishing for the species on the lower Churchill River began as early as 1920 and continued intermittently during the first half of the twentieth century.

² COSEWIC – an independent scientific organization that assesses fish and other species for risk of extinction and is an advisory body to the federal government – does not include the cultural or spiritual significance of a species when assessing whether the species is at risk; nevertheless, COSEWIC does acknowledge that wildlife is more than simply a component of its ecosystem and animals are often imbued with cultural significance which must be recognized (2017).

³ First Nations people used isinglass, a collagen product from sturgeon swim bladders, to make a strong adhesive for attaching drum coverings to frames and for use in painting.

Some of the participants interviewed in the study who worked at the Ilford fish processing plant in the 1970s recall large quantities of Lake Sturgeon being flown in from North Indian Lake, Fidler and Billiard Lakes, and the confluence of the Little Churchill and Churchill Rivers (CNP, 2015).

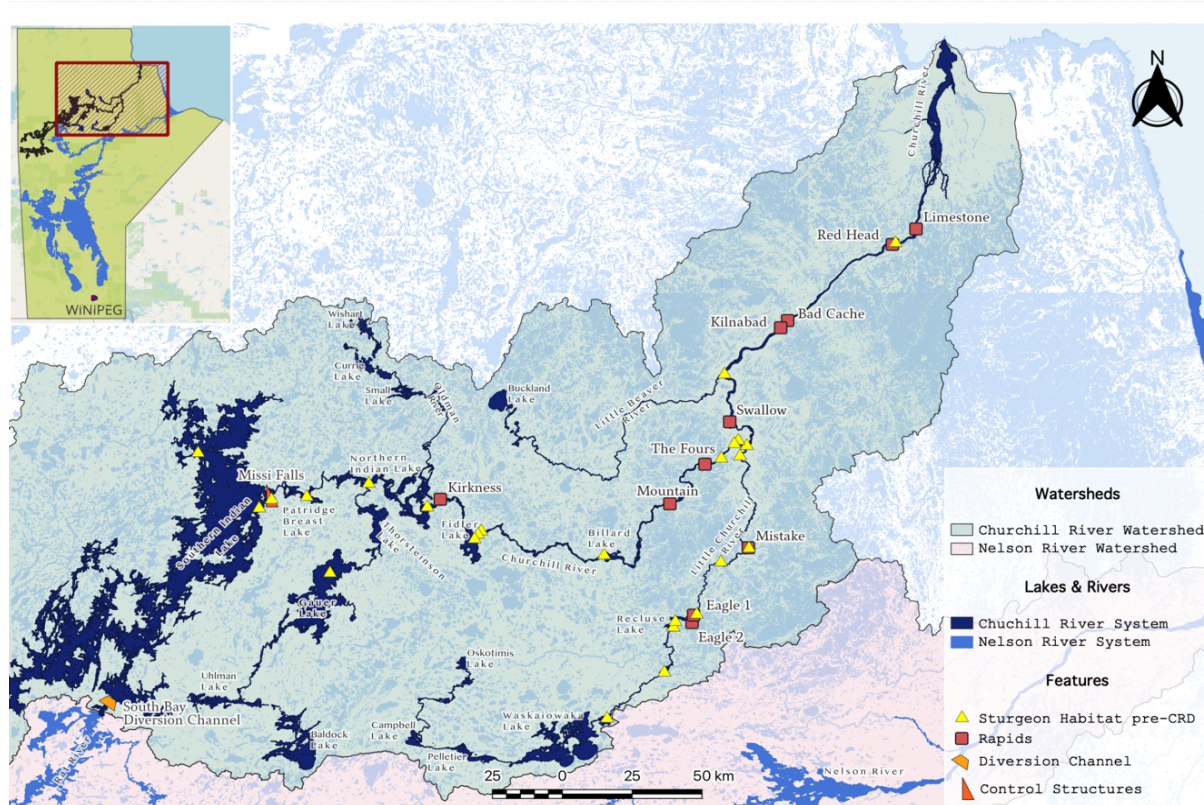


Figure 2: Map of Pre-CRD Lake Sturgeon Habitats along the lower Churchill River according to Indigenous Knowledge gathered by Halket Environmental Consultants.

Elders possess vivid recollections of Lake Sturgeon distribution along the lower Churchill River prior to the implementation of the CRD. Historically, Lake Sturgeon were abundant throughout the entire reach, extending from Missi Falls to the town of Churchill. Insights gathered from recent interviews with Elders and resource users, supplemented by findings from published studies (Edye-Rountree, 2007; Split Lake Cree First Nation, 2016; Tataskweyak Cree Nation, 2012; Cree Nation Partners, 2015), have informed the development of a map identifying key pre-CRD Lake Sturgeon fishing locations (Figure 2). However, Indigenous Knowledge indicates that Lake Sturgeon were regularly caught or observed throughout the river, beyond the specific sites documented below:

- across all the lakes along the river including Southern Indian, Partridge Breast, Northern Indian, Thorsteinson, Billard, and Fidler Lakes;
- at the confluences of tributaries including the Little Churchill and Little Beaver Rivers;

- along tributaries, at Recluse and Waskaiowaka Lakes upstream on the Little Churchill River, and Thorsteinson Lake upstream on the Gauer River; and,
- at the numerous rapids on the Churchill River including Missi Rapids (Falls), Kirkness, Molly, Mountain, the Fours, Swallow and Red Head Rapids, and on the Little Churchill River at Eagle Rapids and Innis-Taylor Rapids during spawning.

Elders recall the splendor of the river and the number of Lake Sturgeon and other fish populations prior to the CRD. With sorrow, they also recall the great decline in Lake Sturgeon numbers and other fish stocks after the CRD:

“While some outsiders claimed that nature had begun to adjust to the unnatural cycles, the Elders continued to notice far more signs of negative effects than positive adjustment. Traditional pursuits continued to be adversely affected along ‘on-system’ waterways and, more and more, hunters, fishermen and trappers had to go further afield in order to harvest game and fish.” (Split Lake Cree, Analysis of Change, 1996).

Other accounts in the “Story of Sturgeon” tell of the tragic loss of a once vibrant ecosystem along the river:

“His initial thought was that sturgeon would not be able to return to their natal or traditional areas, with waterways now blocked due to the very low flows in the Churchill River main stem. Many Elders felt that the Missi Falls Control Structure had done more harm than good... no one really knew what was going on when this early development was under construction. No one realized how widespread the damage from the Missi Falls Control Structure would actually be.⁴ Damage both direct to the river and adjacent lands, and indirect to the culture and traditional patterns of the local users of the river... Many resource users never returned to their beloved river due to "word of mouth" of the widespread devastation and the new dangers the river posed in resource harvesting. Traveling the river and searching for fish or moose, or simply making a life out on that land, has become far too uncertain and dangerous... (per. comm., 2011).”

Another memory of the devastating impact the Missi Falls Control Structure had on Northern Indian Lake:

“[The] fishermen would all notice lots of muddy water and debris surging into the lake' He said he could literally watch the water levels go up in front of his eyes' He mentioned that

⁴ As will be shown in Section 3, DFO scientists predicted the CRD would damage fish and their habitats, although these findings were not entirely shared with First Nations.

August would normally see predictable "hydro floods", where beaver dams would be flooded until nearly frozen up, and [he] said the subsequent erosion, debris and damage to the lakes shoreline was bad and emotional for him (per. comm., 2011).

And on the lower Churchill River:

“Of the respondents interviewed, only four returned to the lower Churchill River area post-CRD to pursue domestic fishing. The reaches previously fished on the Churchill River upstream from the main stem were described as extremely shallow, rocky and with a very swift current it was felt that the new conditions made it an unsuitable habitat for sturgeon (per. comm., 2012)” (Cree Nation Partners, 2015, p.31).

The wide range of flows experienced after the advent of the CRD also caused a commensurately large range in water levels along the river: pre-CRD fluctuations of less than 2 m (6 feet) swelled up to 11 m (36 feet) or more in some places, as they did in 2017 (see Chapter 5, Figure 8), making it nearly impossible for animals to live on the riverbanks year-round:

“Furbearers such as beaver, mink, otter and muskrat cannot survive under these unpredictable fluctuations: because they cannot build or excavate twenty-foot-high houses to accommodate these water level fluctuations, they are frozen out when the water level falls over the winter or drowned if it rises. As a result, these animals are no longer abundant or have disappeared along the river” (per. comm., July 2018).

The range of flows made the river ice dangerous to travel on during the winter:

“The constant fluctuations and unpredictable flows and levels on the lower Churchill make travel on ice or near shore hazardous for both man and machine. Hanging ice⁵ is a constant threat we've seen while skidooing on the Churchill. Shelf or hollow ice is another threat. This happens if the water was high over the freeze-up period and the water drops. The ice that was formed is left suspended on the surrounding boulders on the dewatered river” (per. comm., 2024).

In addition to the dangers of traveling on the ice, resource users have observed very little to no water under the ice:

⁵ Hanging ice occurs when the water below the ice drops and leaves a shelf of ice suspended above the water level.

“We have broken through such hollow ice on our travels on the Churchill River. In a couple instances, while checking the ice conditions with a chisel, we noticed the ice would sound hollow, and when we chopped a hole in the ice there was no water under the ice for a great distance. We've even done this as we walked up to a hole in the ice that turned out to be where the water didn't freeze, as the water made its way down the Churchill River, the main channel. Open ice channels like this ... were where the main flow was at one point when the water was higher, probably during the freeze-up period. We crawled under the ice at a few of these locations and saw nothing but rock boulders, sand, mud and debris. We were sitting, kneeling where water should have been. One location was upriver from the confluence near the Fours Rapids.... [Others were] at Swallow Rapids ... [and] at the mouth of the Little Beaver River” (pers. comm., 2024).

These conditions make traveling, hunting, fishing or trapping anywhere on the lower Churchill River difficult, dangerous and possibly fruitless.

“Traveling anywhere on the Churchill is very treacherous. Places where it should be frozen solid ice isn't. The ice is either hanging or sitting on boulders. Upriver, close to the choke points at the narrower sections, we saw jumble ice⁶ with no water below those sections of jumble ice. Bed rock in a lot of locations were also noted where there should have been feet upon feet of flowing water. We damaged our skidoos traveling on both the Little Churchill and Big Churchill Rivers. We've broken through ice where there was only inches of water to having no water under the ice. This is why we don't travel there by skidoo anymore. The chance of hitting a hole, as described above, or ice shelf or boulder are high and make travel treacherous” (pers. comm., 2024).

First Nation resource users recognize they lost an integral migration corridor for Lake Sturgeon and other fish with the construction of the Missi Falls Control Structure, which imposed a barrier to fish moving upstream from the lower Churchill River into their spawning and overwintering grounds in Southern Indian Lake (SILFA, 2018).

Today, Tataskweyak Elders believe that there is only one surviving recruiting population (i.e., self-sustaining population) of Lake Sturgeon left on the lower Churchill River. Centered at the confluence of the Churchill with the Little Churchill River, elders refer to it as “the Confluence Population”. Our recent Lake Sturgeon eDNA studies (AAE, 2023) confirm this to be true, as did COSEWIC in 2017:

⁶ Jumble ice forms naturally on a river or other flowing bodies of water when the surface layer fractures due to varying flows and currents beneath the ice and then refreezes into an irregular jagged mass. It is characteristic of ice dams occurring on a river.

“In the lower Churchill River downstream of Missi Falls habitat quantity/quality has been severely reduced by the Churchill River Diversion project; however, in this area, the lone recruiting population persists. Further upstream, where no recruiting populations remain, habitat to support all life stages exists in large quantities” (COSEWIC, 2017, p. xi).

Which begs the question: if the habitat does exist upstream, why are Lake Sturgeon absent from it?

3. CRD Regulated Flow Regime and Natural Flow Regime

Flow regulation as part of hydroelectric power development is common on most large rivers throughout Manitoba. Generating stations operate on the Burntwood, Nelson, Saskatchewan and Winnipeg Rivers, along with control structures on the Burntwood and Churchill Rivers. Governing the operation of these dams are licences issued under Manitoba's Water Power Act. The Churchill River Diversion licence regulates the flow into the lower Churchill River.

Under its current licence—set to expire in 2026—Manitoba Hydro is authorized to divert up to 991 m³/s (35,000 cfs) of water during the summer and 963 m³/s (34,000 cfs) in winter from the Churchill River into the Burntwood and Nelson Rivers. Additionally, the licence permits reducing flows in the lower Churchill River to as little as 14 m³/s (500 cfs) in summer and 42.5 m³/s (1,500 cfs) in winter. These allowances represent a potential reduction of up to 98.6% in the river's natural flow, with significant implications for downstream ecosystem integrity.

Flow regulation under the Churchill River Diversion (CRD) is achieved through the Missi Falls Control Structure, located at the head of the lower Churchill River. This structure manages the discharge from Southern Indian Lake into the lower Churchill River (see Figure 3).

Flow regulation poses a dilemma to sustainable resource management: demands on water resources for power must be balanced against environmental interests and the rights of Indigenous people (Alexandra et al., 2023). On the lower Churchill River, power requirements have overridden environmental concerns and Indigenous values, with disastrous results to the fish, wildlife and habitat relied upon by First Nations. As Newbury and Malaher⁷ warned about the under-construction CRD in their 1973 paper entitled, "The Destruction of Manitoba's Last Great River":

"Whether or not this enormous, proposed scheme should become fact is a question that has beleaguered the present government as well as former governments of Manitoba. These governments have doggedly backed Manitoba Hydro's diversion plans, even though there is no doubt that the scheme could create a nightmare for many other human and resource values in the region. The essential question is: Is the power that would be generated really worth more to Manitobans and to Canada than the many values that would be compromised or destroyed? (Newbury and Malaher, 1973, p.3)

⁷ Dr. Newbury is a hydrologist and a retired professor of Civil Engineering and Earth Sciences at the University of Manitoba. Mr. Malaher was director of the Manitoba Wildlife Branch.

Since starting operation in 1976, the Churchill River Diversion (CRD) has significantly altered the flow regime of the lower Churchill River by imposing prolonged periods of artificially low discharge, occasionally punctuated by flood flows exceeding pre-diversion levels. Prior to the CRD, the average flow at Missi Falls was approximately 1,005 m³/s (35,500 cfs). In contrast, diversion to the Nelson River routinely reduces flow at Missi Falls to near the licensed minimums, representing a reduction of up to 98% compared to pre-CRD conditions. This substantial decrease in discharge results in a pronounced contraction of the channel's wetted cross-sectional area, diminished water depth, and reduced flow velocity—hydraulic conditions that would not occur under the natural flow regime.

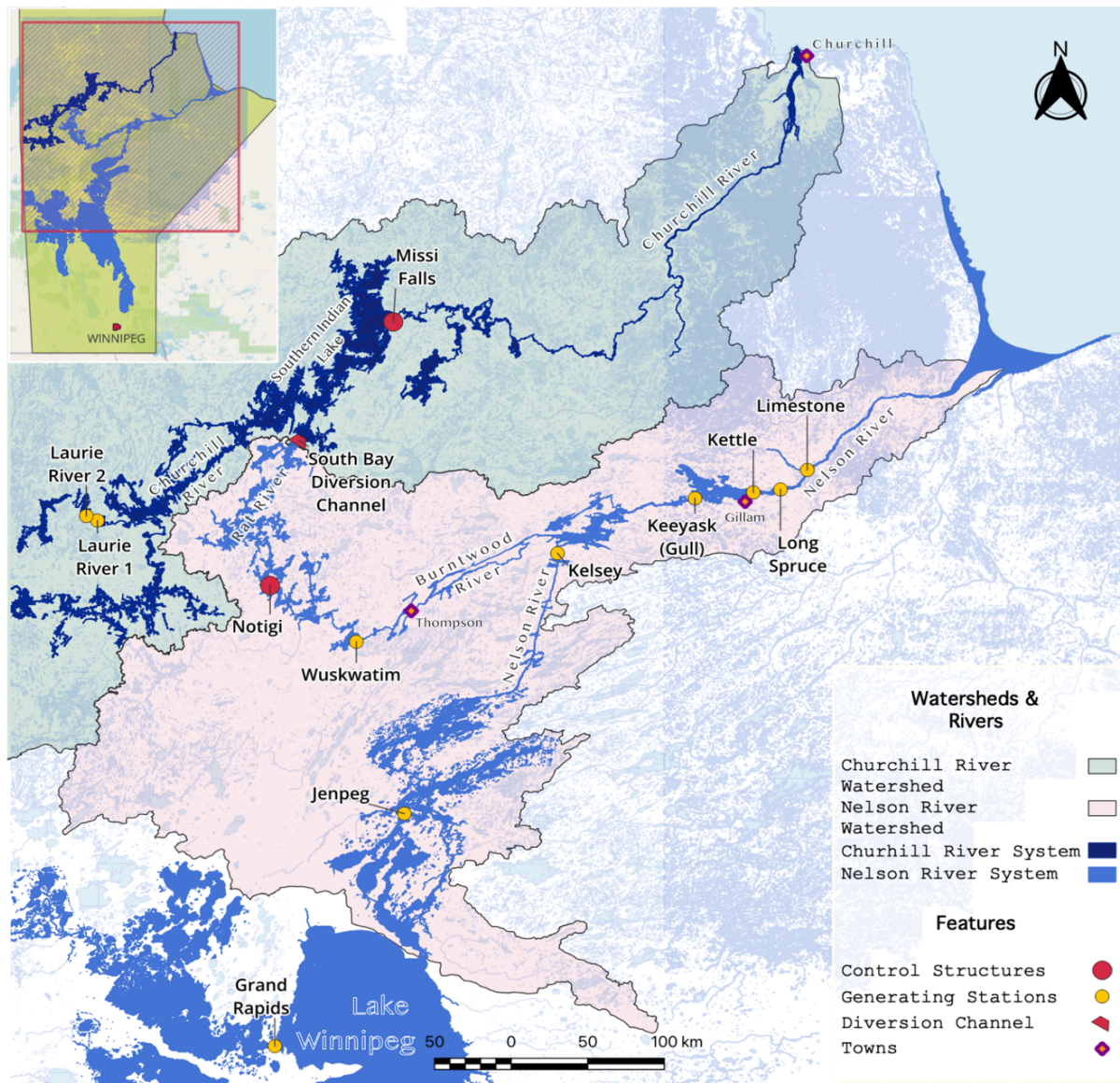


Figure 3: Churchill River Diversion: Hydroelectric Dams and Control Structures.

Missi Falls effectively acts as the safety valve for the complete Manitoba Hydro system of regulated rivers. Under the licence there is no upper limit to the amount of water that Missi Falls can release to the lower Churchill River. To prevent the system from being overwhelmed and flooding communities along the diversion route, the licence allows excess water to be released into the lower Churchill River. For example, in 2017 Manitoba Hydro released a flood of water that reached 3,250 m³/s (113,600 cfs), the highest flow ever recorded on the lower Churchill River.

The installation of gates at the Missi Falls Control Structure introduced a significant barrier to fish movement within the Churchill River system. While fish are able to pass downstream from Southern Indian Lake into the lower Churchill River, upstream passage is obstructed, effectively preventing return migration. Observational evidence following the initiation of CRD operations indicates that large schools of fish became stranded below the control structure. Notably, in 1981—five years after the diversion began—a Fisheries Branch officer documented thousands of Whitefish attempting to return to Southern Indian Lake (Manitoba Hydro, 2015). A report by Barnes and Bodaly (1994) on Whitefish congregating in fall below Missi Falls corroborates this report:

“In 1986 the number of lake whitefish below the dam increased to almost 90,000 in the autumn and most of these fish were physiologically ready to spawn. Morphological comparisons indicated that most of these fish originated from SIL [Southern Indian Lake] and were apparently attempting to return to it. Flows through the dam in this year were high and like those before the diversion of the Churchill River. In 1987 there were fewer fish of SIL origin below the control dam and the numbers decreased over the summer. Discharge through the dam was much lower in 1987 than before diversion. The number of lake whitefish below the dam that appeared to originate from lakes downstream of Missi Falls on the lower Churchill River varied from 2000 to 12000 over the whole study period. These fish may have been attracted to the control dam by local environmental conditions such as cool water temperatures. The effects of the control dam on lake whitefish movements and distribution were not predicted despite extensive pre-project studies. These effects should receive more attention in the prediction of impacts of boreal hydroelectric projects” (Barnes and Bodaly, 1994, p.1).

The impact of the Missi Falls Control Structure on Whitefish spawning, disruption and loss of access to overwintering habitat for Lake Sturgeon has not been adequately investigated by the Province of Manitoba or the Department of Fisheries and Oceans (DFO). This represents a significant ecological concern that warrants focused research and policy attention. In light of these habitat disruptions, the development and implementation of a fish passage strategy should be prioritized to support the recovery and long-term sustainability of affected fish populations.

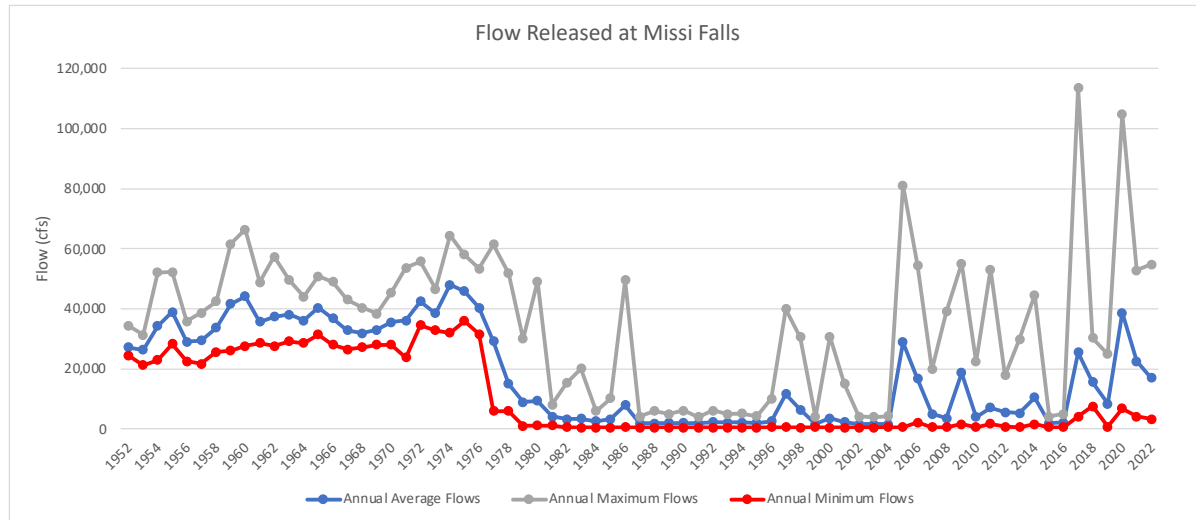


Figure 4: Annual average, minimum and maximum daily flows released to the lower Churchill River at Missi Falls.

Figure 4 illustrates the abrupt change to the river's flow range caused by regulation as shown by the plots of annual maximum, average and minimum daily flows in each year from 1952 to 2022. From 1952 to 1976, the annual average daily flow in the Churchill River was $1005 \text{ m}^3/\text{s}$ (35,500 cfs) (while the maximum flow recorded was $1,877 \text{ m}^3/\text{s}$ (66,300 cfs) and the minimum flow recorded was $603 \text{ m}^3/\text{s}$ (21,300 cfs). The red line shows the precipitous and regularly occurring post-CRD drop in the minimum daily flows to $14 \text{ m}^3/\text{s}$ (500 cfs) in summer or $42 \text{ m}^3/\text{s}$ (1,500 cfs) in winter. The grey line shows the peak average daily flows in each year, which in 2005, 2017, and 2020 surpassed the pre-CRD peak average daily flows.



Figure 5: Images depicting the absence of a functional riparian zone reaching the waterline.

The CRD licence permits a wide range of flow variability on the river, which in turn allows for substantial fluctuations in water levels—exceeding 11 m in some reaches within a single year. These extended periods of low water levels have visibly altered the river’s morphology, most notably in the riparian zone. Once vegetated with trees and other large flora, this zone is now markedly sparse, as illustrated in Figure 5, reflecting the ecological consequences of prolonged hydrological disruption.

The scientific literature consistently identifies the natural flow regime as a foundational element of river health (Poff et al., 1997; Bunn & Arthington, 2002). The seasonal variability and rhythmic patterns inherent in natural hydrology are critical for sustaining ecological integrity. These predictable cycles allow aquatic and riparian ecosystems to adapt and thrive, supporting native biodiversity, facilitating life cycle completion, and maintaining habitat complexity and ecosystem function (Richter et al., 2003; Naiman et al., 2002). Alterations to the natural flow regime, often resulting from dam operations or water withdrawals, can significantly impair river health, leading to declines in native species and disruption of ecological processes (Arthington, 2006; Wohl et al., 2004).

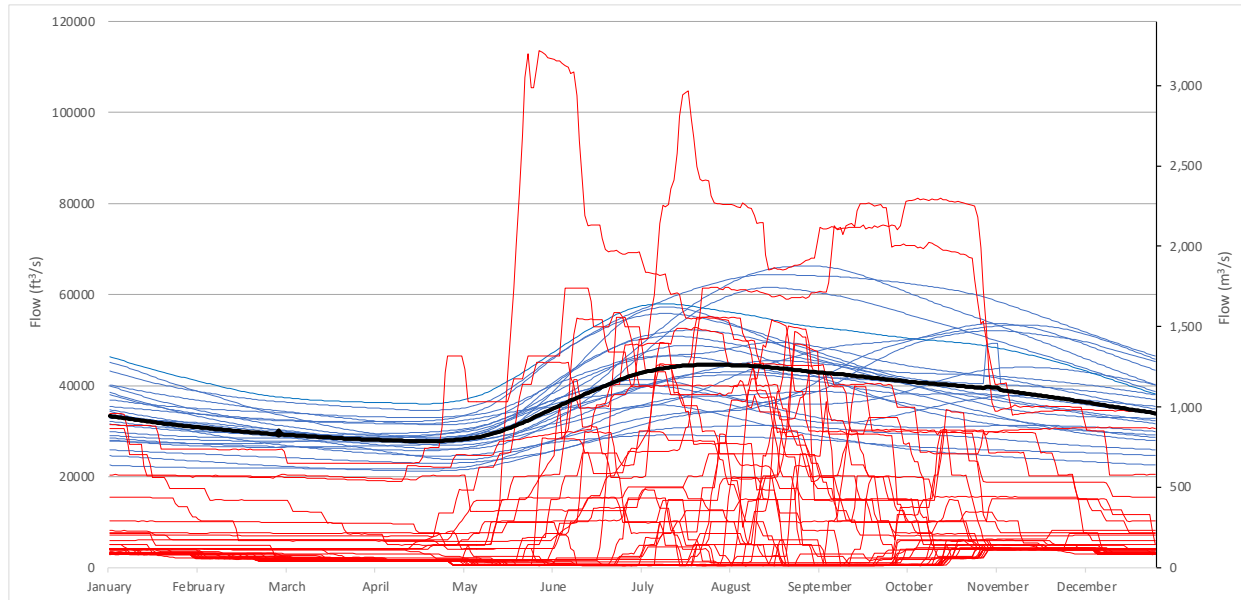


Figure 6: Churchill River annual hydrographs from 1952 to 2022 at Missi Falls. Blue hydrographs represent daily flows from 1952 to 1976; red hydrographs are daily flows from 1977 to 2022 courtesy of Manitoba Hydro. The black hydrograph, the average of daily flows from 1952 to 1974, is the natural flow hydrograph.

As a basis to assess the impacts of flow alteration on fisheries, Fisheries and Oceans Canada suggest a minimum of 20 years of river flow data is required to establish the natural flow regime (FOC, 2013). Annual hydrographs of the flow at Missi Falls from 1952 to 1976 (shown in blue in Figure 6) depict the natural flow regime prior to the CRD, characterized by seasonally responsive and rhythmically consistent flow patterns. In contrast, hydrographs from 1977 to 2022 (shown in red in Figure 6) depict the altered flow regime imposed by Manitoba Hydro, which lacks seasonal coherence and exhibits irregular, block-like fluctuations. This regulated flow regime, in sharp contrast to the natural flow regime, spans a broader range of flow magnitudes and deviates markedly from the predictable patterns observed under natural conditions. Such hydrologic alterations are widely recognized in river science as detrimental to ecological health (Poff & Zimmerman, 2010).

River restoration science practitioners recommend the re-establishment of natural flow regimes as a primary step in rehabilitating degraded rivers (Arthington et al., 2006). However, where full restoration is constrained by competing water demands, the implementation of an environmental flow regime may be necessary. Environmental flows refer to the quantity, timing, and variability of water flows required to sustain freshwater ecosystems and the services they provide to human communities (Acreman & Dunbar, 2004). These flows aim to replicate key aspects of natural flow

regime, supporting habitat formation, water quality, and biological triggers, while accommodating human use and economic considerations.

Implementing an environmental flow regime for the Churchill River is a vital step toward restoring ecological integrity and balancing the cultural priorities of Tataskweyak Cree Nation with the operational priorities of Manitoba Hydro. The distinctive shape of the pre-CRD raster hydrographs, most clearly represented by the average flow hydrograph (shown in black in Figure 6) serves as a foundational reference for revitalizing the river. The remaining challenge lies in determining the extent to which this flow pattern can be shifted downward along the vertical axis of Figure 6 to achieve a balance between maintaining a healthy aquatic ecosystem and meeting the diverse needs of Indigenous communities and hydroelectric operations.

For Tataskweyak Cree Nation, the river is not only a source of water, but a living entity deeply embedded in cultural identity, traditional practices, and community well-being. Adopting an environmental flow regime represents a critical pathway toward healing the river and balancing ecological restoration with the operational needs of Manitoba Hydro. The insights shared through Traditional Knowledge, including observations of Lake Sturgeon decline along the river and the disrupted seasonal patterns post-CRD, are essential to redefining a flow regime. Recognizing the shape of the pre-CRD average flow hydrograph in Figure 6 is the starting point, the challenge now lies in identifying the range of flow levels that the hydrograph must be placed at along the y-axis in Figure 6 to reflect both the seasonal rhythms of a healthy river and the lived experience of those who have depended on its health for generations.

The following chapters synthesize early scientific warnings about the potentially severe ecological consequences of the regulated flow regime proposed by Manitoba, alongside an analysis of the remaining Lake Sturgeon population structure and movement within the Churchill River. Additionally, results from an instream flow model are presented to examine changes in key hydraulic parameters following regulation. Collectively, these components will provide the insight to form an environmental flow regime that not only restores ecological function but also upholds the traditional values of Tataskweyak Cree Nation while accommodating the operational needs of hydroelectric development.

4. Early Scientific Warnings of Drastic Impacts

Before Missi Falls was completed, Newbury and Malaher predicted ominous ecological consequences from the extended low flows briefly punctuated by high flows that would be permitted under the licence:

“Essentially, in the diverted form, the lower 250 miles of the river valley will have only a small flow from local drainage occupying the large unfilled channel. The only appreciable flows will occur in early summer and last only a few weeks when the Southern Indian Lake reservoir is overtopped at the proposed Missi Falls Dam. Whether the smaller flows are maintained during the winter in the large channel without remedial work or extra releases is unknown. **In effect, the abandoned channel aspect will be maintained by short bursts of peak flows forever, and no new natural regime will be allowed to establish. The whole ecology of the Lower Churchill will be drastically changed to some form unique to an artificial regime**” (Newbury and Malaher, 1973, p.9).

During construction of the CRD and under the auspices of the Lake Winnipeg, Churchill and Nelson Rivers Study Board, scientists from Department of Fisheries and Oceans (DFO), universities and consulting engineering firms investigated the probable effects of the CRD on the lower Churchill River. The Study Board assembled their reports (many of which were written in 1974 from studies that began in 1972) into a compendium that was published in 1975, three years after construction began and one year before construction was completed in 1976. The reports warned that the precipitous decline of flow and water levels would alter water quality, turbidity, dissolved oxygen levels in winter, primary and secondary productivity, decline in the fish stocks and lead to the collapse of the commercial fishery.

The reports suggested that the river might even freeze to the bottom during the winter, causing loss of overwintering habitat and widespread mortality of species. For instance, Hecky and Ayles (1974) noted how the sharp decrease to minimum flow would affect the water levels and areas of the lakes:

“Diversion of up to 30,000 cfs [991 m³/s] out of South Bay on Southern Indian Lake into the Rat-Burntwood-Nelson drainage will drastically alter the hydrology of the lower Churchill River and its lakes. The water levels in the lakes are extremely sensitive to discharge as the control sections tend to be narrow and deep. Levels are maintained not so much by a physical horizontal obstruction, e.g. a rock ledge, as by the resistance to flow offered by channel constrictions. If no weirs are constructed to compensate for decreasing the long-term average discharge at Missi Rapids to 8,800 cfs [250 m³/s] from 35,500 cfs [1005 m³/s], the average elevation of these lakes will be decreased by 8.4-10.3 ft [2.5-3.2 m]. Area changes associated

with these lower levels range from a reduction of 39% in Northern Indian Lake to 76% for Fidler under average conditions.... Under minimum discharges allowable by the interim license (500 cfs [14 m³/s] in summer; 1,500 cfs [42 m³/s] in winter), the areas would be reduced another 20-30% of the post-diversion average” (Hecky and Ayles 1974, p.6).

Brown (1974), another DFO scientist, warned of serious changes to the size of the lakes along the lower Churchill River. For example:

“Fidler Lake ... will undergo the most dramatic changes due to its unusual basin form. The large shallow basin will virtually disappear with the 8.4 foot [2.6 m] drawdown. The water area will be reduced by 76% to 3.6 square miles [9.3 km²] and the volume will be reduced by 70%” (Brown,1974, p. 46).

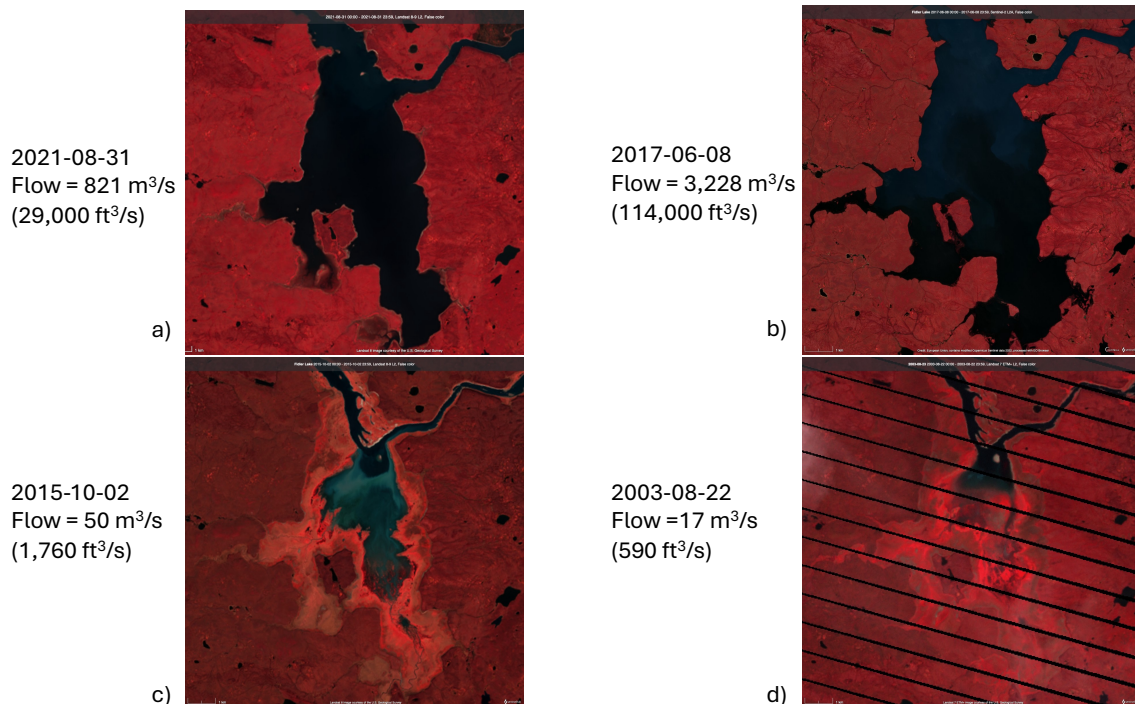


Figure 7: Satellite images of Fidler Lake under different flow conditions. Landsat 8 and Copernicus satellite images courtesy of Sentinel Hub downloaded November 2022.

In Figure 7, the proposed and actual changes caused by the CRD on Fidler Lake are shown by infrared satellite images: Figure 7a captures the lake at a flow level of 821 m³/s (29,000 cfs) on 31st August, 2021, slightly below the average flow level pre-CRD. Figure 7b shows the lake at the highest flow on record in 2017. Figures 7c and 7d shows the lake water levels at flow levels just above the winter and summer regulatory minimums, respectively.

The images of the lake's shrunken extents in Figures 7c and 7d characterize the effect of the regulated minimum flow levels and clearly illustrate the reduction in habitat that was predicted by Brown and other scientists that contributed to the Lake Winnipeg, Nelson and Churchill Rivers Study Board reports. These flow levels have occurred often since 1976 as shown by the post-CRD red hydrographs in Figure 7.

In their 1974 publication, Hecky and Ayles summarize the reports by Ayles et. al. (1974), Brown (1974), Cleugh (1974), Hamilton and McRae (1974), Hausser and Alam (1974), and Patalas and Salki (1974) that analyzed the effect of these flow and water level reductions on the aquatic habitat of the lakes along the lower Churchill River and the cascading effects on productivity and fish. Using a post-CRD average flow of 249 m³/s (8,800 cfs), they predicted a reduction in habitat that would lead to a 66% reduction in primary production and a commensurate reduction in zooplankton. This decrease in food supply would be further worsened by increased turbidity, reduced dissolved oxygen levels in winter, and a shift from the rich Churchill River water quality to a more spartan quality present in local lakes. They go on to warn that the impact on fishing would be dire:

“Changes in the whole fish populations are likely to have the greatest impact. Catch per unit effort will decline and fishing effort on the lakes could cease entirely because of this” (Hecky and Ayles, 1974, p.10).

Even more alarming are their predictions that most fish would be killed under the CRD licence minimums of 14 m³/s (500 cfs) in summer and 42 m³/s (1,500 cfs) in winter. Not only do they predict a further 20-30% reduction in the lakes' sizes with commensurate reductions in productivity, but they also warn:

“With the accompanying complete loss of aquatic habitat due to lower water levels and reduced area, it is clear that there will be a drastic reduction in the potential fish production of the lower Churchill lakes. The recently completed ice study by LaSalle Hydraulic Laboratories indicates that under minimum allowable discharges (present interim license) the lower Churchill River downstream of the lakes will freeze solid **obliterating most aquatic life**. The greater water level fluctuations on the lakes and the probability of freeze-out in the lower river stretches entail the risk of probable violations of the Federal Fisheries Act (which has been communicated to the Study Board through previous correspondence...)” (Hecky and Ayles, 1974, p.11, emphasis added).

The LaSalle Report referenced by Hecky and Ayles was commissioned by the Study Board to investigate Newbury's 1971 claim that on the lower section of the river below Fidler Lake,

“...flow stoppages due to icing in the lower reaches during midwinter under minimum release conditions may be encountered” (Newbury, 1971, p.ii).

Newbury recognized that during winter low-flow conditions, water would disperse across the broad riverbed of the Churchill River, a channel originally shaped by a much higher natural flow regime. This dispersion over the wide bed would result in shallow flow depths, increasing the risk of the river freezing to its bed.

The LaSalle Report authored by Hausser and Alam (1974) and included in the Study Board’s reports, agrees with Newbury’s conclusion. The authors investigated three levels of flow, 113, 283, and 991 m³/s (4,000, 10,000, and 35,000 cfs) and for these levels calculated the amount of flow that would be used up to form ice during freeze-up under average conditions. For the three levels they determined that 62, 70, and 108 m³/s, (2,200, 2,500 and 3,800 cfs) would be used to form ice. Their calculations show that at the 113 m³/s level over half of the water flow would be used up to make ice.

Hausser and Alam (1974) added a caveat to their calculations:

“these figures are based on mean values of monthly temperature and average ice cover progression. In reality, there could be short periods of extreme low temperature when the flow reduction might increase by another 50%. In terms of present discharge, these flow reductions represent only a small percentage of the total flow, but when compared to post-diversion conditions, they are practically equal to the flow remaining in the river” (p.23).

Considering Newbury’s guidance alongside the calculations presented by Hausser and Alam, it is evident that the winter minimum flow of 42 m³/s (1,500 cfs) falls significantly below the 113 m³/s (4,000 cfs) threshold identified in their analysis—by nearly a factor of three. This discrepancy raises serious concerns about the potential for the river to freeze to its bed during severe winter conditions. To date, this aspect of hydrological impact and its implications for ecosystem health does not appear to have been formally investigated by either Manitoba or Manitoba Hydro.

4.1. Absence of Environmental Review by Federal or Provincial Authorities

Indeed, the flow regime allowed under the CRD’s licence has resulted in significantly lowered flows and water levels interspersed with historically high floods. The effect of this regulated regime has never been the subject of an environmental review by Federal or Provincial authorities.

The Lake Winnipeg, Churchill and Nelson Rivers Study Board Reports study’s predictions regarding the impacts of operating under the CRD minimum flow conditions during winter (ie

freezing to the bed) went unheeded. Concerns about Federal/Provincial inaction were being raised by Newbury and Malaher as early as 1973:

“Lack of basic data and the failure to undertake in-depth quantitative resource studies, in spite of pressure to do so, is a serious indictment of both governments concerned. Even the one detailed study, (The Lake Winnipeg, Churchill and Nelson Rivers Study Board Reports) which deals with the effects on resources only, will not be completed in time to influence the course of events in any significant way.

Newbury and Mahar were particularly scathing about decision makers’ choice not to wait for or heed the Study Board reports’ recommendations before making decisions:

“...[i]n a final move of arrogance, on December 8, 1973, the Minister responsible for granting a license to Manitoba Hydro for the diversion, the Hon. Sydney Green, announced that no license would be required, no hearings into the issue would be held, and no compensation to native communities would be necessary as it had been decided in Cabinet to simply grant permission to Manitoba Hydro to proceed immediately by an Order in Council. The legality of this move is now in question” (Newbury and Malaher, 1973, p.16).

To this day no independent scientific study has investigated the Study Board’s 1974 predictions of the CRD’s effects on the lower Churchill River.

The Province of Manitoba and Manitoba Hydro conducted what was described as a Regional Cumulative Effects Assessment (RCEA) of Hydroelectric Developments on the Churchill, Burntwood and Nelson River System in 2015. However, the Manitoba Clean Environment Commission (2018), Tataskweyak Cree Nation (2018) and other First Nations expressed significant concerns about the RCEA. For example, that the RCEA:

- did not follow accepted guidelines for a cumulative effect assessment; (p.49, 67)
- conducted their study without asking First Nations what was important to them; (p. 29, 49, 50)
- did not include shorelines and benthic organisms in its choice of Regional Study Components (RSC) or valuable environmental components (p. 49, 64). This is problematic because:
 - the exclusion of shoreline habitats (riparian and littoral) as an RSC resulted in a large gap in the regional analysis. There are 25,000 kilometers of shoreline in the study area and the shorelines along rivers and lakes are by far the most impacted habitats by hydroelectric development; (p. 49, 61, 64)
 - benthic organisms are near the base of the food chain and their population abundance and composition can be highly affected by changes in the water

regime and have a cascading effect on other components of the ecosystem that depend on them (such as fish, waterfowl and furbearers); (p. 61)

- did not include in the temporal scope a firm pre-development baseline and therefore the changes to present day conditions were not fully investigated nor did it attempt to cast impacts into the future, a departure from best practice; (p. 49, 51) and
- limited the assessment to effects of hydroelectric development. It is standard practice to incorporate the effects of other developments or actions that may cause effects or interact with effects caused by hydroelectric development. (p. 61) (RCEA Review, 2015, p. 49 to 67).

Pertinent to the overwintering conditions for fish on the river, the RCEA makes only one brief reference to the CRD's effect on flow under ice conditions:

“Churchill River Diversion resulted in reduced winter discharge and reduced water velocities along the lower Churchill River. Lower velocities result in increased area with ice cover. There would also be less ice accumulation by shoving and thickening at the leading edge which typically occurs downstream of fast flowing open water river sections” (RCEA, 2015, section 4.3 p. 61).

This passage does not address the amount of the flow required for ice formation, how thick ice will be under different flow conditions, whether ice will freeze to the riverbed at the regulated low flow level, and how the previous 40 years of the CRD's winter operations have impacted fish and their habitats.

In fact, the RCEA offers limited analysis of the aquatic ecosystem concerns raised by Tataskweyak Cree Nation and the Lake Winnipeg, Churchill and Nelson Rivers Study Board. Despite early warnings from scientists affiliated with the Study Board—who cautioned about the significant ecological risks associated with river diversion—there is scant evidence that federal or provincial governments have taken substantive action in response to those concerns.

5. Lake Sturgeon on the Lower Churchill River

Lake Sturgeon are an ancient and historically successful species of fish, having survived relatively unchanged for 200 million years. But the modern world has upset this stability, and Lake Sturgeon are now classified as either Endangered, Threatened or Of Special Concern throughout much of their range in Canada (COSEWIC, 2017) Tataskweyak resource users are well aware of this status:

“We know from our perspective that the Lake Sturgeon are endangered and that they need our protection. We know that they need Canada’s protection” (pers comm, 2017).

Elders describe the Lake Sturgeon as a “picky” fish, selective about their environment and easily “grossed out” or “disgusted” if the water is not clean. Scientific studies have yielded similar findings, suggesting that Lake Sturgeon are highly vulnerable to habitat alteration and over-fishing because of their unique qualities. (COSEWIC, 2006).

Lake Sturgeon have a life cycle roughly like that of humans, characterized by slow growth, late maturity, and periodic spawning. One of the largest freshwater fish, they can reach lengths of up to 3 meters and weights of up to 180 kilograms, though estimates vary. Tataskweyak resource users share that they have caught Lake Sturgeon as large as 30 kg in the post-CRD period but noted that overall size is declining.

5.1. Habitat Requirements and Lake Sturgeon Life Stages

According to Fisheries and Oceans Canada, Lake Sturgeon require a variety of habitats during their life cycle, from spawning to foraging. The quality and availability of these habitats, which include spawning, nursery, feeding, and overwintering areas, are crucial for the species' survival. In addition, an open connection between habitats is paramount for the survival of this species as they are known to migrate over great distances to find their preferred habitat requirements (Bemis and Kynard, 1997).

Lake Sturgeon habitat is determined by several physical factors such as water flow, depth and velocity, substrate composition, water quality and water temperature. A description of the habitat requirements for each life history stage is provided below.

5.1.1. Spawning

Lake Sturgeon spawn in late spring (May to June) when water temperatures rise above 10°C, typically around 13°C to 18°C. Their longest migration, repeated throughout the Lake Sturgeon’s life, occurs to find suitable spawning habitat. They prefer fast flowing water (0.5-2.0 m/s) like

rapids or the base of a waterfall, with depths of 0.3 to 6.0 meters. Spawning areas ideally have coarse substrate bottoms such as gravel or cobble, to which the eggs will adhere to once they are fertilized and be protected from predators (Priegel and Wirth, 1974; Becker, 1983; LaHaye et al., 1992; Auer and Baker, 2002; Bruch and Binkowski, 2002; Kerr et al., 2011; COSEWIC, 2017).

5.1.2. Larval stage

Lake Sturgeon eggs hatch about a week after fertilization. Larval Lake Sturgeon remain in the vicinity of the spawning grounds, using the spaces between boulders, cobble, and gravel as cover from predators. Once they have completely absorbed their yolk-sac, they will begin to drift downstream in search of nursery habitat. Lake Sturgeon larvae may drift between 11 and 26 km to find suitable habitat consisting of shallower water depths, lower flow currents, and relatively vegetation-free substrates such as sand or clay that will offer an abundant food supply (Kempinger, 1996; Smith and King, 2005). Larval lake sturgeon transform into young of the year (fry) about 14 to 30 days after hatching, once they absorb their yolk sac and begin to feed independently. At this stage, they begin to look like miniature Lake Sturgeon and start to swim and forage actively. Larval and young-of-the-year Lake Sturgeon, prefer relatively shallow water (2 to 5 m deep) with low flow (1 to 2 cm/s) but in the Winnipeg and Nelson Rivers, larval and young-of-the-year Lake Sturgeon have also been captured in moderate to deep-water habitats (Manitoba Hydro, 2016).

5.1.3. Juveniles

Juveniles, usually 1 to 10 years of age, show preference for deeper habitat than adults, with low-velocity water (Benson et al., 2005; Kempinger, 1996). Juvenile Lake Sturgeon feed on benthic invertebrates such as insect larvae, leeches, isopods, molluscs, and crustaceans (Harkness, 1980; Wallus, 1990; Chiasson et al., 1997), and generally remain in close contact with bottom substrates (Kempinger, 1996). Juveniles have been found over a wide range of substrates including gravel, cobble, and boulders in addition to clay and sand, which they seem to prefer (Barth et al., 2009; Department of Fisheries and Oceans, 2010). Habitat type seems to be of less importance than the availability of benthic prey items (Chiasson et al., 1997; Kerr et al., 2011). After reaching sub-adult sizes, habitat preferences change to deeper, higher velocity waters, much closer to the habitat preferences of adults.

5.1.4. Adults

Adult Lake Sturgeon prefer fine substrates comprised mainly of sand, silt and clay which are associated with higher densities of prey items (Kerr et al., 2011). They feed mainly on benthic invertebrates including insect larvae, freshwater shrimp, leeches, snails, clams and small fish (Harkness, 1923; Harkness and Dymond, 1961; Priegel and Wirth, 1974). Adults prefer deep, slow-moving water at temperatures of less than 20°C (Harkness and Dymond, 1961; Priegel and Wirth, 1974; Kerr et al., 2011). Adult Lake Sturgeon can adapt to various habitat conditions more readily than juveniles (DFO, 2010), and are more likely to move through narrow waterways than

juveniles (MBH, 2016). While adult Lake Sturgeon may have larger movement patterns than juveniles, there is evidence that adults show site fidelity within their home ranges and throughout seasonal migrations (MBH, 2016).

5.1.5. Overwintering

In autumn, as temperatures start to drop, both juvenile and adult Lake Sturgeon exhibit reduced movement and seek overwintering habitat (COSEWIC, 2017). Lake Sturgeon generally move in a downstream direction (Caroffino et al., 2009), out of river and into lake environments (COSEWIC, 2017). During the winter, Lake Sturgeon in the lower Churchill River exhibit minimal movement, spending most of their time in deep, slow-moving pools where they can remain relatively inactive. These areas are often characterized by stable conditions, such as slow currents and higher dissolved oxygen levels, which are essential for the Lake Sturgeon's survival during the cold months when metabolic rates are low, and food is scarce.

While there are knowledge gaps regarding the overwintering requirements for young-of-the-year and juvenile fish, overwinter habitat is generally downstream of spawning and nursery habitat and it has been suggested that juvenile overwintering habitats are similar to those of adults (Caroffino et al., 2009). During the winter months, Lake Sturgeon are more sedentary and do not feed as actively as other times of the year. Groups of Lake Sturgeon can often be found sharing overwintering habitat (Kerr et al., 2011). When spring arrives and water temperatures begin to increase, Lake Sturgeon exhibit strong fidelity to spawning sites, often returning to their natal streams to spawn (Wilson et al., 2004).

5.2. Lake Sturgeon eDNA, Population and Movement Studies

Between 2018 and 2023, a series of environmental DNA (eDNA), population, and movement studies were conducted on Lake Sturgeon within the lower Churchill River and its associated lakes. eDNA analyses performed in 2022 and 2023 corroborate conclusions drawn by COSEWIC and Tataskweyak Cree Nation, indicating the presence of a single recruiting Lake Sturgeon population in the river system. This population, referred to as the “confluence population,” is located at the confluence with the Little Churchill River (AAE, 2023b). Collectively, the data lend support to assertions by Tataskweyak and other First Nations that the CRD has led to the extirpation of Lake Sturgeon from most of the river.

Population estimates derived from capture–recapture statistical models suggest that the confluence population comprises approximately 2,100 adult individuals (AAE, 2025a).

In 2019, 47 adult Lake Sturgeon, each exceeding 800 mm in fork length, were surgically implanted with acoustic transmitters to monitor movement patterns. Over the subsequent five-year period,

tracking efforts were maintained with minimal loss, as only one individual was lost due to accidental mortality.

Movement data indicate that Lake Sturgeon spatial distribution during the open-water season is influenced by flow and water level conditions. Specifically, higher discharge and elevated water levels were associated with broader movement ranges (Table 1). These findings are consistent with observations reported by resource users from Tataskweyak Cree Nation.

Table 1: Movements of the Confluence Population from 2019 to 2023.

Confluence Population - Annual Movement upstream into Little Churchill River										
Year	2019		2020		2021		2022		2023	
Minimum Flow (cfs)	27-Jun	3,270	24-Apr	6,850	14-Apr	19,705	13-Apr	4,200	26-Apr	2,210
Maximum Flow (cfs)	09-Sep	27,935	27-Jul	98,175	25-Jul	57,200	24-Jun	65,685	21-Jul	7,840
Percentage of population that moved up the Little Churchill River	15%		90%		93%		75%		51%	

During the high-flow years of 2020, 2021, and 2022—when hydrological conditions more closely approximated the river’s natural flow regime—most Lake Sturgeon from the confluence population migrated from the Churchill River into the Little Churchill River during early summer for spawning and foraging activities. By autumn, typically between September and November, all individuals returned to overwinter within the main stem of the Churchill River, specifically occupying a 40 km reach between Mountain Rapids and Swallow Rapids.

In contrast, the year 2019 was characterized by persistently low flows and the absence of a spring flood, resulting in the confinement of most tagged Lake Sturgeon to the Churchill River throughout the year. Telemetry data presented in AAE’s 2025b Movement Report indicate that by the time peak flows occurred in early September, individuals had already initiated their transition to overwintering habitats and did not migrate into tributary systems. Similarly, in 2023, another low-flow year, only approximately half of the tagged population entered the Little Churchill River, while the remainder remained within the main stem.

These findings confirm that low spring and summer flows restrict Lake Sturgeon movement, primarily due to the elevated position of the Little Churchill River’s outlet channel relative to the Churchill River’s bed. Conversely, when flow levels remain elevated throughout the year, as observed in 2020–2022, Lake Sturgeon are able to access tributaries for spawning and foraging. Importantly, regardless of seasonal movement patterns, all individuals consistently return to the lower Churchill River to overwinter.

6. Instream Flow Model

Hydrological parameters such as water depth, flow velocity, substrate or bed material composition, and channel connectivity are critical determinants of Lake Sturgeon survival across all life stages (COSEWIC, 2017; Kerr et al., 2014). The Committee on the Status of Endangered Wildlife in Canada (COSEWIC) has identified that the lower Churchill River possesses suitable substrate conditions throughout its length, which support the habitat requirements of Lake Sturgeon (COSEWIC, 2017). Natural connectivity within the river system is governed by the interaction between flow regimes and substrate characteristics, both of which influence habitat accessibility and ecological integrity.

Given the altered flow patterns following the implementation of the CRD, it is reasonable to infer that changes in key hydraulic parameters, specifically wetted cross-sectional area, water depth, average velocity, and channel connectivity, are likely the primary drivers influencing habitat suitability for Lake Sturgeon in this system. Despite the availability of appropriate substrate, these hydrodynamic alterations may significantly constrain the species' ability to access and utilize critical habitats.

To investigate these changes, an instream flow model was applied to the lower Churchill River with two primary objectives: (1) to evaluate the extent to which Lake Sturgeon habitat has been modified under the regulated flow conditions imposed by the CRD licensing framework; and (2) to identify an environmental flow regime that supports the long-term viability of Lake Sturgeon populations while accommodating Manitoba Hydro's operational requirements for water diversion to the Burntwood and Nelson Rivers.

An instream flow model is a computational tool designed to simulate water movement within a river. It can also evaluate how hydraulic parameters like wetted cross-sectional area, water depth and water velocity, respond to changes in flow. Once properly calibrated, the model can reliably estimate these key hydraulic parameters for different levels of flow along a river. From comparisons of the changes between the hydraulic parameters at pre-CRD flow levels and post-CRD flow levels, the effects of regulation on habitat can be assessed.

For the lower Churchill River, the model was applied to simulate pre-CRD conditions, the natural flow regime, and the post-CRD regulated regime. This allows for a detailed examination of how flow alterations affect Lake Sturgeon habitat and supports the development of an environmental flow strategy that can align Tataskweyak Cree Nation's cultural and ecological priorities with Manitoba Hydro's operational requirements.

The successful development of our model of Southern Indian Lake (Halket, 2018), located directly upstream of the lower Churchill River, provided our approach for constructing an instream flow model tailored to the lower Churchill system. The Southern Indian Lake model simulates hydrodynamic and thermal responses under varying flow and climate scenarios, both prior to and following the CRD. This dual capability was essential for the lower Churchill River model, which needed to compare pre-CRD conditions, when Lake Sturgeon populations were robust, with the current regulated regime, where only one recruiting population remains. Indigenous knowledge and assessments by COSEWIC (2017) have highlighted the severe decline in Lake Sturgeon populations under post-CRD conditions, underscoring the importance of modelling these changes to inform environmental flow strategies.

Recall that the 440 km long lower Churchill River stretches from Southern Indian Lake to the town of Churchill. Major lakes along the river's upper reach are Partridge Breast, Northern Indian, Fidler, and Billard. Except for the long channel between Fidler and Billard Lakes, the upstream lakes are joined by relatively short channels, whereas below Billard Lake, there is a long channel to Hudson Bay. The channels are composed of pool and riffle reaches (alternating deep and shallow sections of a river) interspersed with hydraulic features such as rapids and shoots. As Hecky and Ayles (1974) point out, water levels on the river are extremely sensitive to changes in discharge because the flow is controlled mainly by the roughness of its bed and banks rather than being controlled by barriers to flow.

The modeling team initially considered a two-dimensional (2-D) approach but found it impractical due to the study's scope, limited data, and long computation time. Since 2-D models are better suited for floodplain simulations, and this study focuses on low, in-channel flows, a one-dimensional (1-D) model was selected for its efficiency and suitability.

DHI's MIKE modeling suite includes MIKE Hydro River (1-D), MIKE 21 (2-D), and MIKE 3 (3-D), offering flexible integration across dimensions. These models can be linked and interchanged, allowing targeted refinement where needed; for example, inserting 2-D or 3-D sections into a primarily 1-D river model. The 3-D model developed for Southern Indian Lake, which defines the upstream boundary, can be connected to the 1-D model of the lower Churchill River to create a comprehensive simulation of the Churchill River system in Manitoba.

Hydro River is a 1-D river model that can model a variety of tasks related to river hydraulics, water quality, flooding, forecasting, and navigation as well as catchment dynamics and runoff. MIKE Hydro River requires cross-section data that describe the morphology of the river channel combined with topographic data which describe the river's floodplain. Boundary conditions describe water inputs and outputs. Hydrographs created with external tools feed into the model as time series files to describe the boundaries.

The development of any MIKE Hydro River model is a multi-step process of collecting available hydrologic and meteorologic data, building a sufficiently detailed model framework, calibrating the model by running known inputs and comparing model results against measurements of water levels and/or flow rates at hydrometric stations or periodic observations at other locations. Following satisfactory model calibration, the model can simulate the effect of various flow conditions on water velocities, depths, wetted areas, and temperatures. The basic inputs required by MIKE Hydro River are channel cross-sections along the river in the form of x, y and z coordinates (where x and y denote the horizontal position and z corresponds to elevation or depth); the roughness coefficients of the channel and the floodplain, representing the friction encountered by flowing water; and boundary conditions which provide the upstream and tributary flows.

6.1. Model Setup – Surveying Cross-Sections

The lower Churchill River remains one of the least studied areas affected by hydroelectric development in Manitoba. Building an instream flow model required spatial data to define the framework and flow or water level data to establish boundary conditions. A key focus of the study was identifying data gaps and acquiring missing information—most notably, the absence of surveyed cross-section data for the river channel.

Cross-sections are the basis for forming a computer model of the river channel, and the basis for habitat analysis. A cross-section is a slice through the river channel perpendicular to the flow direction which depicts the channel's bed, banks and the water surface. It shows the habitable underwater area, as shown in Appendix A.

A total of sixty-seven cross-sections were surveyed along the lower Churchill River (Fig. 8). In 2019, the survey covered the stretch from Billard Lake to Swallow Rapids. Due to unsafe high-water conditions in 2020 and 2021, efforts shifted to the Little Churchill River and its confluence with the Churchill River. The survey resumed in 2022, focusing on the reach between Northern Indian and Fidler Lake, and was completed in 2023 with the section from Fidler to Billard Lake.

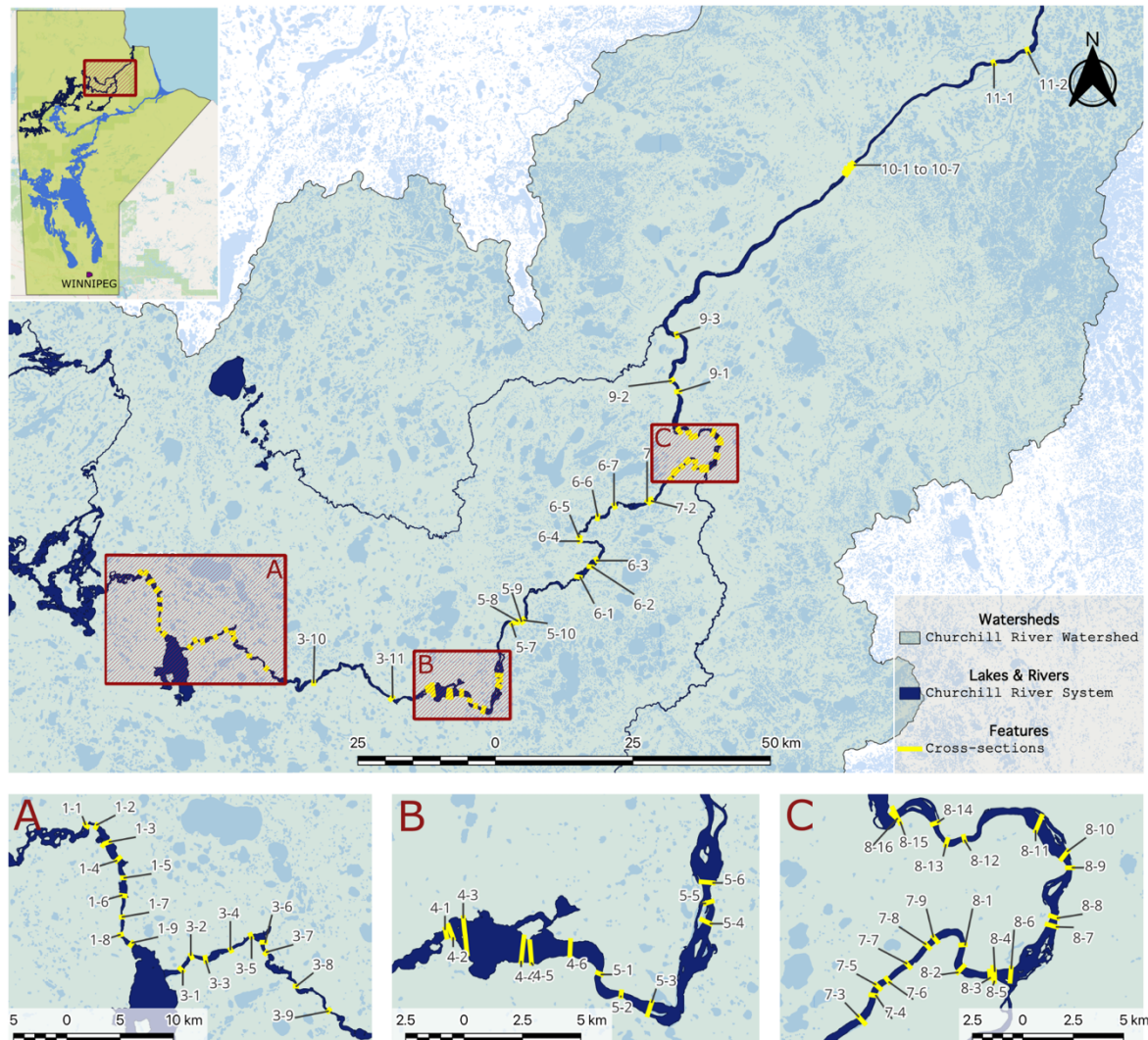


Figure 8: Cross-sections on the lower Churchill River.

The cross-sectional surveys spanning from Northern Indian Lake to downstream of Swallow Rapids were completed over five field seasons, conducted under challenging and often arduous conditions. Surveys were conducted over two-week periods, necessitated by the logistical challenges of navigating the lower Churchill River while transporting survey and camping equipment.

Survey crews and equipment were flown by float plane from Thompson to remote sections of the Churchill River. Teams traveled by canoe, skiff, or raft, requiring lightweight, portable gear for portaging around rapids. Operations depended on safe weather and river conditions, and without powered boats, crews conducted surveys while drifting downstream, then paddled back upstream each evening to retrieve GPS equipment.

Each cross-section was surveyed from the left to right bank (looking downstream), capturing both above- and below-water features. Above-water morphology was recorded using a Trimble GNSS RTK 4 system, while three to five acoustic transects with a Biosonics MX Echosounder defined the underwater profile. Post-processing was completed in Winnipeg to finalize the cross-section data.

Coordinates (x and y) followed NAD83, Zone 15N, and elevations (z) were referenced to CGVD2013a using Natural Resources Canada's Precise Point Positioning System.

The cross-sections and substrate composition of the riverbed are presented in Appendix A. The cross-section's number and date surveyed are shown at the top of the figures. The cross-sections vary in width from a few hundred metres to over a thousand metres. Also, presented below each cross-section are a description of the percentage makeup of materials forming the riverbed, as illustrated in Figure 1-1. Interestingly, these descriptions of the materials forming the riverbed confirm COSEWIC's (2017) assertion that suitable substrate exists along the lower Churchill River to support all life stages of Sturgeon.

6.2. Model Framework, Boundary Conditions

The model's framework merged three different spatial data sets:

- A Digital Elevation Model (DEM), based on a 90-metre grid of the Churchill River from Missi Falls to the Port of Churchill, available from Manitoba Growth, Enterprise and Trade's Geoscience web page: this dataset was imported into ARCGIS and trimmed to define the river's floodplain in x,y,z coordinates.
- Lake bathymetries were framed based on the Study Board's comprehensive set of maps depicting the bathymetries (underwater depths) of Partridge Breast, Northern Indian and Fidler Lakes, and the Coordinated Aquatics Monitoring Program (CAMP) also has recent maps that depict the bathymetries of Northern Indian and Billard Lakes.
- Sixty-seven cross-sections, surveyed from Kirkness Rapids, near the outlet of Northern Indian Lake, to just downstream of the Swallow Rapids.

The framework was formed by importing the DEM and the bathymetry data for the lakes into ARCGIS and exporting the combined DEM to MIKE in x, y, z coordinates. The surveyed cross-section data in x, y, z format were set into the DEM to form a digital representation of the river's channel between Missi Falls and Swallow Rapids.

The model used four boundaries: an upstream boundary, a downstream boundary, and two open boundaries at the confluences with Gauer and Little Churchill Rivers. We assumed that the riverbed/water boundary, which is the interplay between groundwater inflows, and seepage to groundwater would offset each other and could be switched off for all model runs.

Descriptions of the boundary conditions for model simulations were as follows:

- The upstream boundary at Missi Falls was defined by a time series of average daily flows over the calibration period (2016 to 2017) provided by Manitoba Hydro and by constant daily flows to simulate flow levels of interest.
- The downstream boundary, located downstream of Swallow Rapids at distance of approximately 300 km from the upstream boundary, used Manning's equation to estimate the water surface elevation (stage) at the downstream end of the model based on the computed flow and a specified friction slope (energy grade line slope).
- Two tributary boundaries at the Gauer and Little Churchill Rivers' confluences were defined by their constant average annual daily discharges of 52 and 33.5 m³/s, respectively.

6.3. Model Calibration and Error Analysis

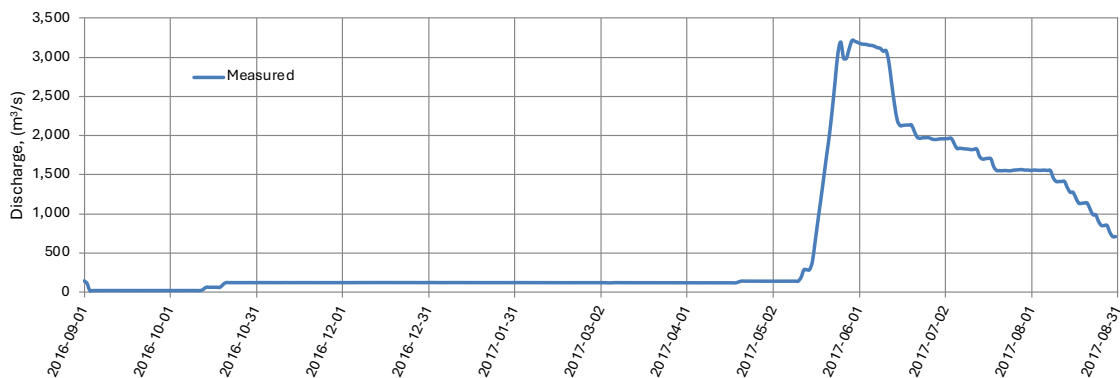


Figure 9: Hydrograph of flow released from Missi Falls Control Structure to the lower Churchill River from September 2016 to August 31, 2017.

The model was calibrated by comparing model simulation results to measured water levels and flows recorded at hydrometric stations downstream of Missi Falls from September 1, 2016 to August 31, 2017 (Fig. 9). This period, chosen because it covers the greatest range of flows measured on the river, encompasses three distinct flow periods:

- a summer low flow period from September to October 2016, with flows averaging 20 m³/s (706 cfs) released at Missi Falls;
- a winter low flow period from September to May, with an average flow between approximately 20 and 120 m³/s (700 to 4,500 cfs) released at Missi Falls; and,
- a spring freshet flow period from May to July 2017, including the highest flow on record with a peak flow of 3,256 m³/s (115,000 cfs).

Model calibration initially involved iterative simulations in which the roughness coefficient, representing the frictional resistance exerted by the channel bed and banks, was adjusted to achieve the best fit between simulated and observed hydrographs. Calibration was based on three hydrographs derived from flow and water level data collected at two hydrometric stations downstream of Missi Falls: the Water Survey of Canada station below Fidler Lake (Station No. 06FB001), which provided both flow and water level measurements, and the Manitoba Hydro station on Billard Lake (Station No. 06FB702), which recorded water levels only. A Manning's roughness coefficient (n) value of 0.036 yielded the optimal agreement between simulated and measured hydrographs at both locations.

Comparison of the plots of measured versus simulated hydrographs in Figures 10, 11 and 12 show a very close fit. Also, of note in Figures 10 and 12 are the ranges between the high and low water levels recorded at these two stations: greater than 10 m and 6 m, respectively.

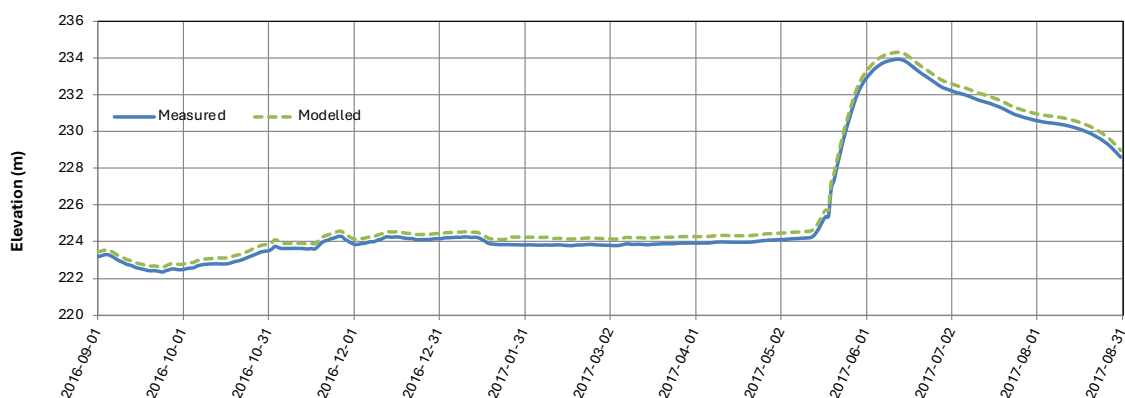


Figure 10: Comparison of measured and modelled water levels at the Water Survey of Canada Hydrometric Station below Fidler Lake.

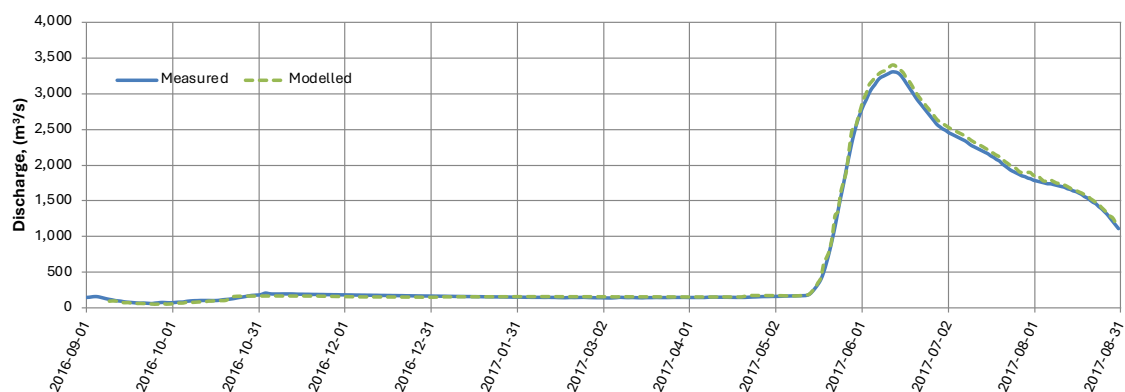


Figure 11: Comparison of measured and modelled flows at the Water Survey of Canada Hydrometric Station below Fidler Lake.

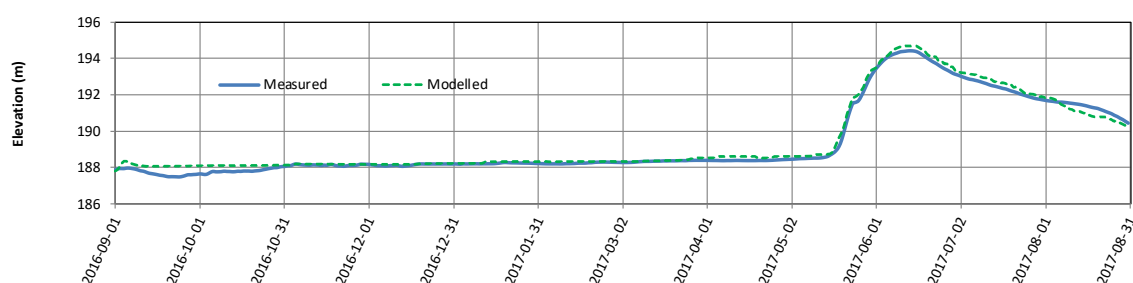


Figure 12: Comparison of measured and modelled water levels at the Manitoba Hydro Hydrometric Station at Billard Lake.

To measure the accuracy of the model's predictions, four error statistics were calculated:

- Nash–Sutcliffe Efficiency (NSE), a widely used and reliable statistic for assessing the goodness of fit for hydrologic models. The index determines the relative magnitude of the modelled variance compared to the measured variance; the closer to one the better the fit.
- Difference between means in metres, which measures the absolute difference between mean values for the two groups of data.
- Bias, which indicates whether the model results consistently underestimate or overestimate the measured data. A positive value suggests the model will underestimate the measured data, and a negative value indicates the model overestimates it.
- Root Mean Square Error (RMSE), which is the standard deviation of the residuals (prediction errors). RMSE provides an estimation of how well the model predicts the target value (accuracy).

Values for these statistics for the calibrated model runs are shown in Table 2 below.

Table 2: Goodness of Fit Statistics.

	Fidler Lake Water Levels	Fidler Lake Flows	Billard Lake Water Levels
Nash-Sutcliffe Efficiency	0.99	1	0.99
Difference between Means	-0.33 m	-13.27 m ³ /s	-0.14 m
Bias	-0.15%	-1.70%	-0.07%
Root Mean Square Error	0.33 m	37.0 m ³ /s	0.23 m

The model's good performance is strongly supported by its high Nash-Sutcliffe Efficiency (NSE) scores, 0.99 for water levels at Fidler and Billard Lakes, and a perfect match for flow at Fidler Lake, indicating an excellent fit between predicted and observed data. Root Mean Square Error

(RMSE) values further confirm the model's accuracy across a wide range of flow conditions, from seasonal lows to record highs.

All three comparisons show a negative bias, meaning the model slightly overestimates water levels and flows: by 0.33 m for Fidler Lake levels, 13.2 m³/s for Fidler Lake flows, and 0.14 m for Billard Lake levels. As a result, it will also overestimate related hydraulic parameters like wetted cross-sectional area, average depth and flow velocity. This suggests that actual low-flow conditions may be more severe than the model predicts.

6.4. Results

The model simulates changes in three hydraulic parameters that describe physical habitat conditions along the river for different flows under open-water conditions:

- wetted cross-sectional area which represents the amount of underwater habitat;
- average depth of the channel, because different life stages require different depths; and
- average velocity, because different life stages require flows of different speeds.

Hydraulic parameters were calculated at each of the 67 surveyed cross-sections along the lower Churchill River for seven distinct flow levels. Three of these levels of flow, 708, 991, and 1,400 m³/s (25,000, 35,000, and 50,000 cfs), are representative of the natural flow regime prior to the implementation of the CRD. The remaining four flows—14, 42, 142, and 425 m³/s (500, 1,500, 5,000, and 15,000 cfs) reflect regulated flow conditions typical under the CRD operational framework. As illustrated in the historical flow record (Figures 4 and 5), these lower post-CRD flow levels did not occur prior to the diversion and represent a significant departure from the river's natural hydrological patterns.

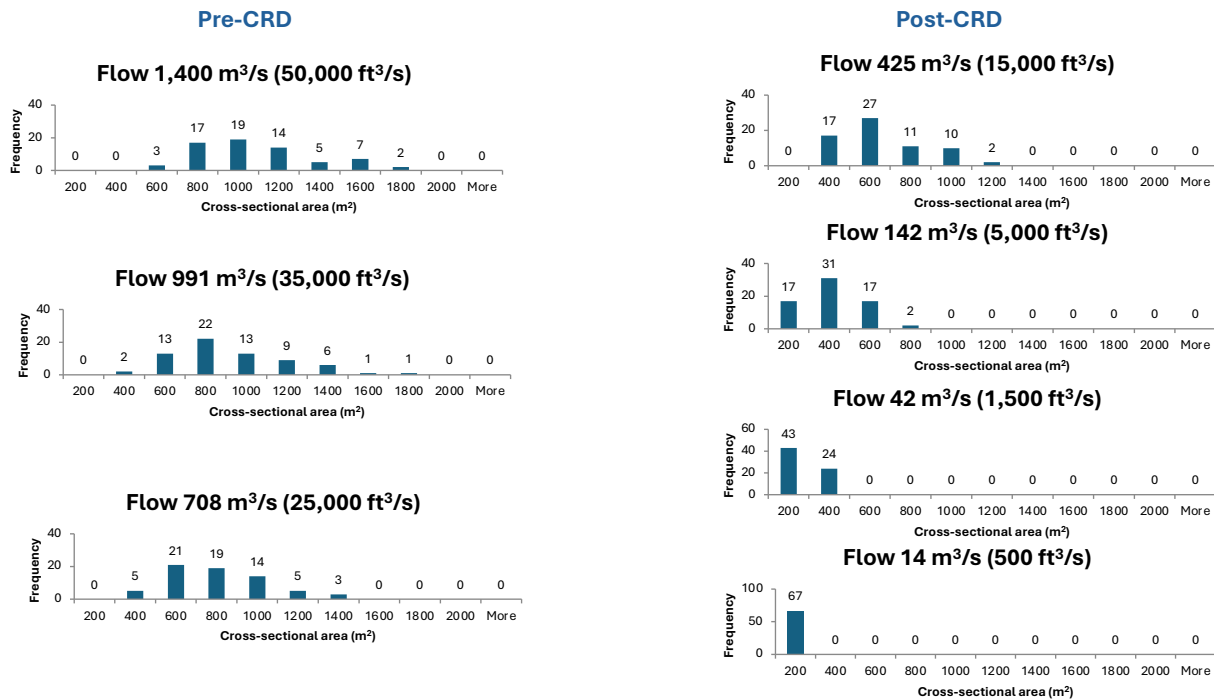


Figure 13: Comparison of the wetted cross-sectional area distributions across flow levels.

Figure 13 highlights the contrast between pre- and post-CRD hydraulic conditions by illustrating the relationship between flow levels and wetted cross-sectional area, an effective measure of available underwater habitat for Lake Sturgeon. The left side of the figure represents typical flow conditions prior to CRD, while the right side depicts flow levels that have only occurred since the onset of regulation, and which are significantly lower than what naturally occurred.

A clear distinction is apparent when comparing the natural flow regime distributions on the left side to the regulated flow distributions on the right-side in Figure 13. Flow levels above 708 m³/s (25,000 cfs) exhibit normally distributed (bell-shaped) wetted areas, while flows below that threshold show skewed distributions. The normal distributions are indicative of pre-CRD river conditions, an environment where Sturgeon populations once thrived. In contrast, the skewed distributions reflect flow levels only experienced after regulation, highlighting the reduction in habitat diversity caused by the CRD.

For the pre-CRD minimum flow of 708 m³/s (25,000 cfs), not one of the cross-sections has a wetted area below 200 m² with five out of 67 cross-sections having wetted areas between 200 and 400 m². The spread of the 67 wetted areas conforms to a normal distribution (bell shaped) stretching from 200 m² out to 1400 m². Higher flows also show the same pattern, albeit with a higher range of wetted areas. These three graphs describe the habitat conditions of the natural flow regime that nurtured a healthy Lake Sturgeon population throughout the entire lower

Churchill River—the normally distributed wetted areas show the range in area and diversity of habitats that Lake Sturgeon require for all life stages. On the other hand, the post-CRD flow graphs do not exhibit normal distributions. They are skewed and squished into a lower range of wetted areas until, at the minimum regulated flow levels of 42 m³/s (winter) and 14 m³/s (summer), they occupy the narrow ranges below 400 and 200 m², respectively. The figure illustrates the severe reduction in wetted areas caused by the low flows after regulation, and the consequent loss in area and diversity of underwater habitats for Lake Sturgeon and, in fact, for all aquatic species.

Recall that our model only simulates flow under open-water conditions. Under ice-forming conditions, the reduction in habitat will be even worse, further endangering survival. In the LaSalle Hydraulic Laboratory report, which investigated the effect of ice formation on the river, Hausser and Alam calculated how much of the flow would be lost to the formation of ice during the initial freeze-up period. For a release of 113 m³/s (4000 cfs) at Missi Falls, they calculated that 62 m³/s (2,200 cfs) or over half of the flow would be used to form ice. Further, they warned that the calculation was made for average conditions and could be up to 50% more for colder than average periods. They go on to warn that:

“Under extreme minimum discharge of 1,500 cfs [42 m³/s] at Missi Falls it is anticipated that major part of the flow section in the river downstream would be frozen fast. Only a small discharge about 100 cfs [3 m³/s] would most probably reach downstream, flowing through the thalweg and loose ice blocks” (Hausser and Alam, 1974, p.iv).

The results of our open-water model can be used with Hauser and Alam’s calculations to infer conditions under ice formation in the following manner: for a flow of 142 m³/s (5,000 cfs), we can expect that about half of the flow would be used to form ice, resulting in cross-sectional areas more akin to the 42 m³/s (1,500 cfs) open-water flow condition.

According to Hausser and Alam, flows of 42 m³/s (1,500 cfs) in early winter may cause sections of the river to freeze completely to its bed, a condition more severe than our model predicts for open-water scenarios at the same flow. Resource users from Tataskweyak Cree Nation have observed areas beneath the ice where no water remains during minimum releases from Missi Falls. Under these conditions, aquatic life experiences extreme stress due to restricted space and movement, leading to habitat degradation and increased risk of mortality for species such as Lake Sturgeon.

A model capable of simulating ice formation was not employed due to the absence of ice thickness data required for calibration. Consequently, any results generated under such conditions would be speculative and lack empirical support.

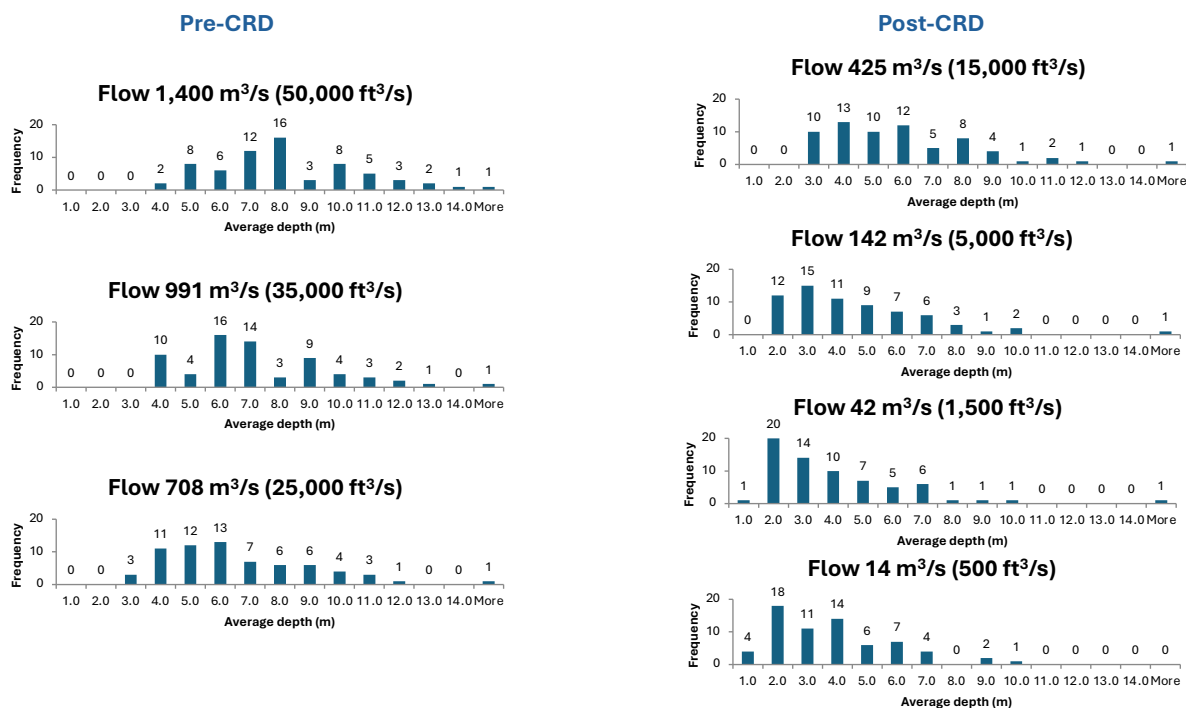


Figure 14: Comparison of average depth distributions across flow levels.

Figures 14 and 15 highlight the shift in average water depth and average flow velocity between pre- and post-CRD conditions. Lake Sturgeon depend on a range of flow speeds and depths throughout their life cycle: fast, shallow water for spawning, and slow, deep pools for adult foraging and overwintering. Juveniles prefer intermediate habitats. While adults typically occupy depths of 5–10 m for foraging, studies from the Winnipeg River show they are also found in water deeper than 10 m (COSEWIC, 2006; Lacho et al., 2021), consistent with the natural flow regime depths illustrated in Figure 14.

In Figure 14, average depths on the left-hand side vary between 3 m to more than 14 m and follow the pattern of a normal distribution. These depth distributions represent the natural flow regime present before the operation of the CRD. On the other hand, as flow levels on the right-side of the figure decrease, there is an absence of deep-water habitats as the histograms become more skewed.

Deep-water overwintering habitats are critical to the survival of Lake Sturgeon in the lower Churchill River. For adult Lake Sturgeon in Canada, overwintering typically occurs in deep, slow-moving pools within lakes and large rivers, with depths ranging from less than 10 m to over 40 m. Within the study area, only one cross-section, located at the base of Mountain Rapids west of the confluence, exceeds 14 m in depth. Under the 425 m³/s (15,000 cfs) post-CRD flow scenario, only three additional cross-sections exceed 10 m in depth. In contrast, for flow scenarios of 142 m³/s

(5,000 cfs) and lower, no cross-sections deeper than 10 m were identified. This progressive reduction in deep-water habitat availability under declining flow conditions is a matter of significant ecological concern, as it likely contributes to the observed absence of Lake Sturgeon from most reaches of the river, except for the confluence reach where the remaining deep-water refuge exists.

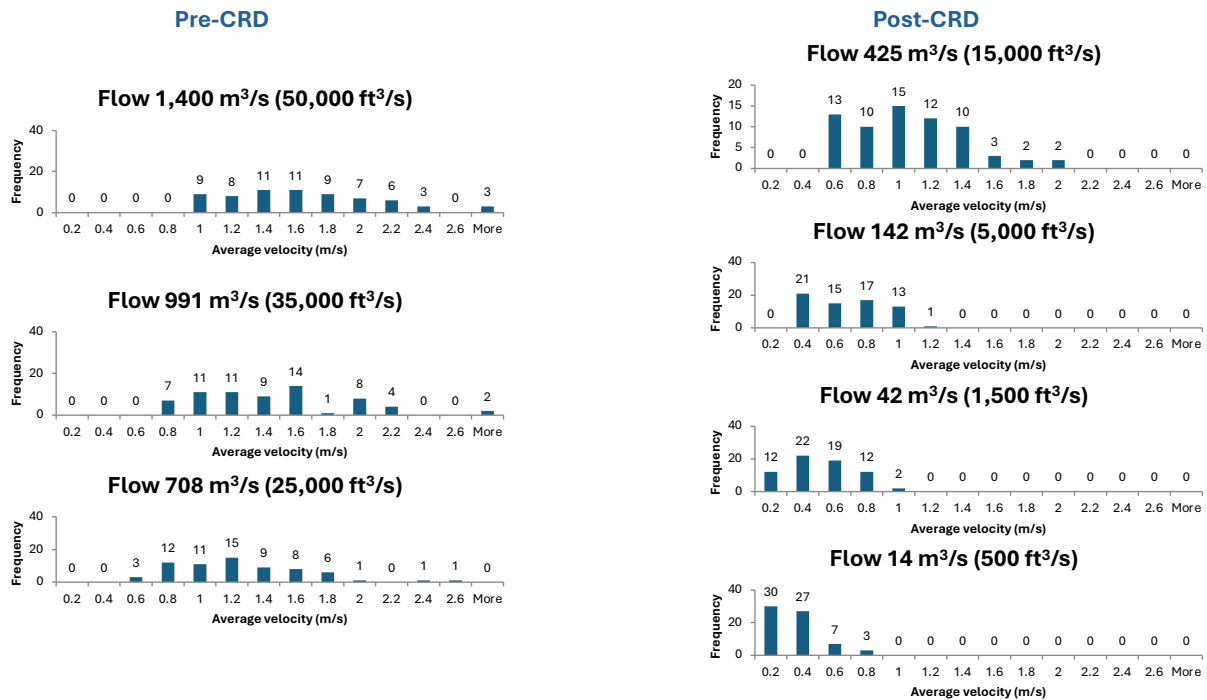


Figure 15: Comparison of average velocity distributions across flow levels.

Figure 15 illustrates the contrast in average flow velocity distributions under natural and regulated flow regimes. The left-hand panels, representing pre-CRD conditions, show that average velocities follow normal distributions. In contrast, the right-hand panels reveal increasingly skewed distributions as flow levels decline under regulated conditions. At a flow of 14 m³/s (500 cfs), all surveyed cross-sections exhibit average velocities below 0.8 m/s. At 42 m³/s (1,500 cfs), 65 out of 67 cross-sections remain below this threshold. This pattern diverges markedly from pre-CRD conditions, where flows of 708 m³/s (25,000 cfs) and 991 m³/s (35,000 cfs) produced average velocities ranging from 0.4 to 2.2 m/s, with median values between 1.0 to 1.2 m/s and 1.4 to 1.6 m/s, respectively. As outlined in Section 4.1.1, Lake Sturgeon require flow velocities between 0.5 and 2.0 m/s for successful spawning, a range well represented in the natural flow regime histograms. In contrast, the histograms corresponding to lower regulated flows indicate a substantial reduction in fast-water habitat availability, with such conditions becoming nearly absent at licensed minimum flows, posing a significant risk to Lake Sturgeon spawning success. The observed shift in both the range and distribution shape of average velocities between 425 m³/s (15,000 cfs) and 708 m³/s (25,000 cfs) suggests that the lower threshold for an ecologically sustainable flow regime lies within this interval.

A comprehensive assessment of the impact of licensed low flow conditions on habitat availability requires an understanding of both the frequency and duration of these flows. To this end, post-CRD daily flow data provided by Manitoba Hydro were analyzed to determine the number of years in which low flows occurred, as well as the number of days per year during which such conditions persisted. As illustrated in Figure 16, the results confirm that the lower Churchill River has been subjected to prolonged periods of extremely low flow throughout the 46 years of CRD operation, spanning from 1976 to 2022.

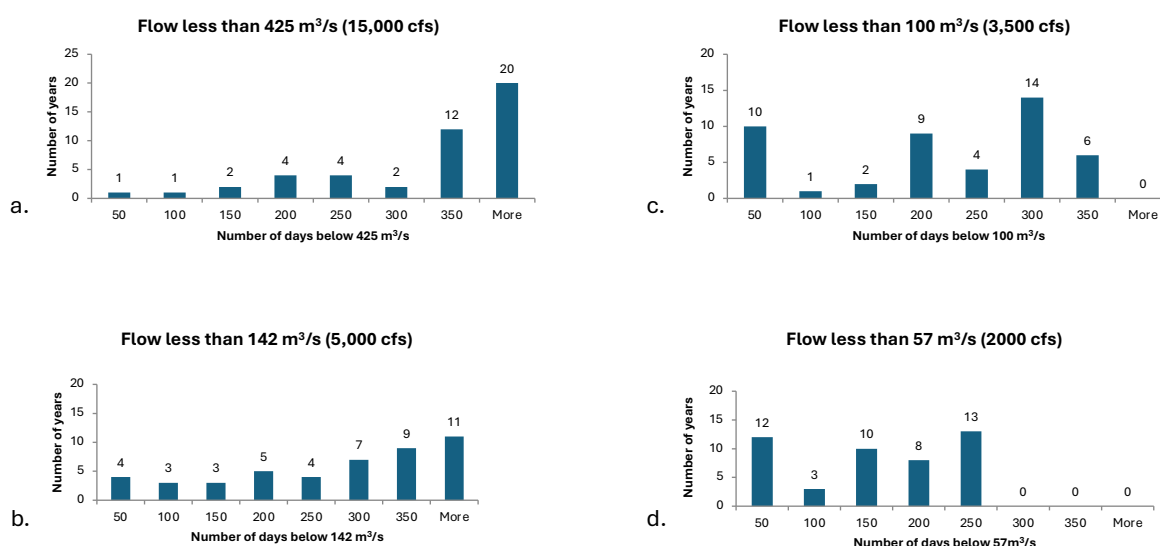


Figure 16: Histograms showing the number of years and days in a year where flows remained below 425 m³/s (15,000 cfs), 142 m³/s (5,000 cfs), 100 m³/s (3,500 cfs) and 57 m³/s (2,000 cfs).

As illustrated in Figure 16a, flow conditions below 425 m³/s (15,000 cfs) occurred for more than 350 days—virtually the entire year—in 20 of the 46 years between 1976 and 2022. Additionally, flows below this threshold persisted for over 300 days in 32 of those years. Flows below 142 m³/s (5,000 cfs), shown in Figure 16b, were recorded for nearly the entire year in 11 years and persisted for more than 200 days annually in 31 of the 46 years.

As shown in Table 1, Lake Sturgeon were confined to the main channel of the river throughout the year when flows dropped below 93 m³/s (3,270 cfs). From Figure 16c, flows below 100 m³/s (3,500 cfs) occurred for at least 200 days per year in 24 of the 46 years, while flows below 57 m³/s (2,000 cfs), Figure 16d, persisted for more than 200 days annually in 13 years.

These findings, supported by statistical analyses and the annual hydrographs drawn in Figure 6, underscore the frequency and duration of low flow conditions since the implementation of flow regulation. Such persistent low flows have likely contributed to significant ecological impacts, particularly on Lake Sturgeon and other aquatic and riparian species in the lower Churchill River. Although COSEWIC (2017) notes that Lake Sturgeon have been extirpated from upstream reaches, extensive areas of suitable substrate remain. However, the viability of these habitats depends on a consistent and sufficient water supply, corresponding to the lowest flow level at which key hydraulic parameters; wetted cross-sectional area, average depth, and flow velocity, exhibit normal distribution characteristics necessary to support Lake Sturgeon across all life stages.

The restoration of Lake Sturgeon populations in the lower Churchill River necessitates an increase in the minimum flow levels currently permitted under the CRD licensing framework, to maintain a consistent and ecologically supportive water supply. River rehabilitation science emphasizes the importance of aligning regulated flow regimes with natural hydrological patterns. Model simulations indicate that under pre-CRD conditions, key hydraulic parameters, wetted cross-sectional area, average depth, and flow velocity exhibited normal distributions, which were associated with healthy and widely distributed Lake Sturgeon populations.

As flow levels decline below 708 m³/s (25,000 cfs), the distributions of these parameters become increasingly skewed, with the degree of distortion intensifying at lower flows. This shift has emerged exclusively under regulated conditions. Both Indigenous Knowledge and COSEWIC (2017) have documented the disappearance of Lake Sturgeon from the river following regulation, a decline that was anticipated by researchers prior to the implementation of the CRD. The flow threshold at which hydraulic parameter distributions transition from normal (indicating a diverse range of habitats) to skewed (reflecting a limited range of habitats) delineates the boundary of the lowest natural flow regime. This regime represents an environmental flow level capable of sustaining ecological integrity and supporting the cultural values of Tataskweyak Cree Nation, while accommodating Manitoba Hydro's operational requirements. This threshold lies between 425 and 708 m³/s (15,000 and 25,000 cfs).

7. Conclusion and Recommendations

First Nations communities along the lower Churchill River have long observed the decline of Lake Sturgeon and other aquatic species following the implementation of the CRD. Their traditional ecological knowledge, such as the role of beaver dams in shaping aquatic and terrestrial habitats, has consistently warned of the CRD's harmful impact on the river's ecosystems.

Scientific predictions made prior to the CRD's development clearly anticipated severe habitat degradation due to reduced flow. Despite these warnings and repeated objections from First Nations, government authorities proceeded without conducting long-term ecological assessments. To this day, no formal environmental impact study has been undertaken.

This report integrates multiple lines of inquiry, including Indigenous Knowledge, river rehabilitation science, pre-CRD impact assessments, Lake Sturgeon population and movement studies, and instream flow modeling, to investigate the decline of Lake Sturgeon in the Churchill River. Collectively, these findings provide clear evidence of the population's collapse and inform recommendations for establishing an environmental flow regime to support recovery.

Elders recall that Lake Sturgeon once thrived throughout the Churchill River. Today, only a remnant population remains, confined to a 40 km stretch between Mountain Rapids and Swallow Rapids. eDNA surveys confirm their extirpation from all other reaches, supporting COSEWIC's 2017 conclusion that this is the last recruiting population on the river.

Movement studies show that when flows mirror the natural flow regime, Lake Sturgeon migrate into tributaries like the Little Churchill and Beaver Rivers to spawn and forage. But under regulated flow conditions, these inlets become too shallow, trapping sturgeon in the main stem year-round with limited habitat and food.

Our instream flow model reveals stark differences in hydraulic conditions before and after the CRD. As flows drop to the minimums allowed under the CRD licence, key habitat variables, wetted cross-sectional area, average water depth, and average flow velocity, become skewed, indicating loss of diversity of habitats and unsustainable conditions. These impacts are even more severe under ice formation, where habitat loss is compounded.

The evidence is clear: prolonged low flows under the CRD licence have led to the collapse of Lake Sturgeon populations and degradation of their habitats. Yet, the riverbed substrate remains intact and could support recovery if flows are restored. As Hecky and Harper (1974) noted, increased flow through Missi Falls would enhance downstream productivity.

To reverse this ecological decline, minimum flow levels must be raised. The following recommendations outline a path toward recovery.

Recommendation 1: Establish an environmental flow regime.

We recommend implementing an environmental flow regime that sets a minimum discharge of 425 m³/s (15,000 cfs) in early spring, gradually increasing to 708 m³/s (25,000 cfs) by August 1, and tapering back to 425 m³/s by the following spring. This seasonal pattern would approximate the river's natural hydrograph (see Figure 17).

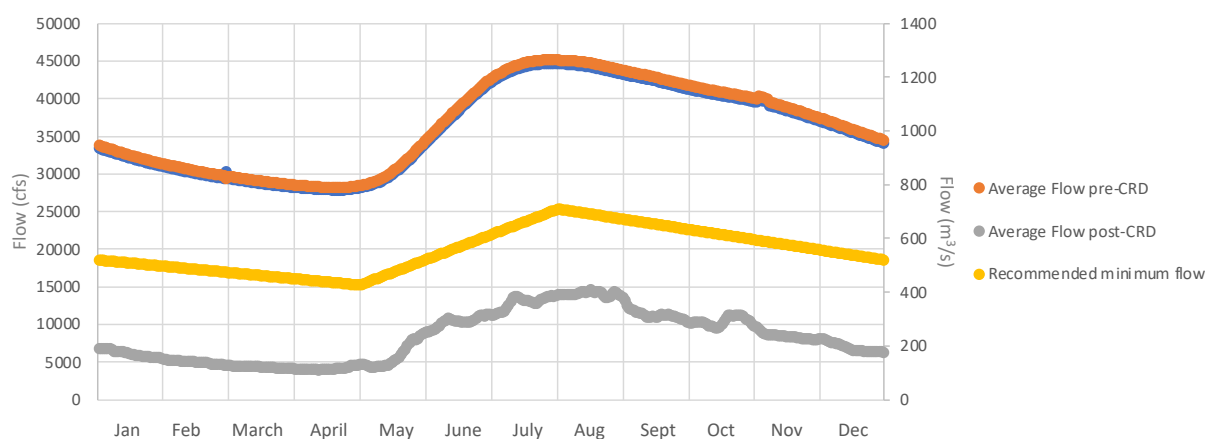


Figure 17: Recommended minimum average daily flow release from Missi Falls Control Structure.

Such a regime is expected to restore key habitat conditions—wetted areas exceeding 200 m², depths greater than 2 m, and flow velocities above 0.4 m/s—which historically supported Lake Sturgeon populations prior to the Churchill River Diversion (CRD). Furthermore, based on the winter ice formation analysis by Hausser and Alam, maintaining these flows would reduce the risk of the river freezing to its bed, thereby safeguarding overwintering aquatic life.

It is also essential that any deviations from the natural flow pattern be introduced gradually to minimize ecological disruption.

Recommendation 2: Implement a Targeted Lake Sturgeon Restocking Program.

Following the restoration of an environmental flow regime, natural habitat recovery is expected to support the gradual return of Lake Sturgeon. However, to accelerate recolonization upstream of Mountain Rapids, we recommend initiating a targeted restocking program. As confirmed by COSEWIC (2017), the Lake Sturgeon population in the Churchill River is genetically distinct;

therefore, restocking efforts should exclusively utilize individuals from the remnant local population to preserve genetic integrity and support long-term recovery.

Recommendation 3: Construct a Fishway at Missi Falls.

Missi Falls currently acts as a barrier to fish movement into Southern Indian Lake. We recommend constructing a fishway to re-establish connectivity for Lake Sturgeon and other native species such as whitefish, enabling access to historic spawning and foraging habitats. In addition to enhancing population resilience across the Churchill River system, the fishway would provide access to essential deep-water overwintering habitat for Lake Sturgeon.

Recommendation 4: Fund Long-Term Monitoring.

We recommend that Canada and Manitoba provide sustained funding for a long-term monitoring program led by Tataskweyak Cree Nation. This program would track Lake Sturgeon population trends and assess the effectiveness of flow restoration, restocking, and fish passage measures over time.

Recommendation 5: Review Licence Conditions Every Five Years.

We recommend conducting a formal review of any future authorization under the federal Fisheries Act and licence under the Manitoba Water Power Act every five years. These reviews should assess whether the revised flow regime is effectively contributing to Lake Sturgeon recovery and allow for adjustments based on long-term monitoring results.

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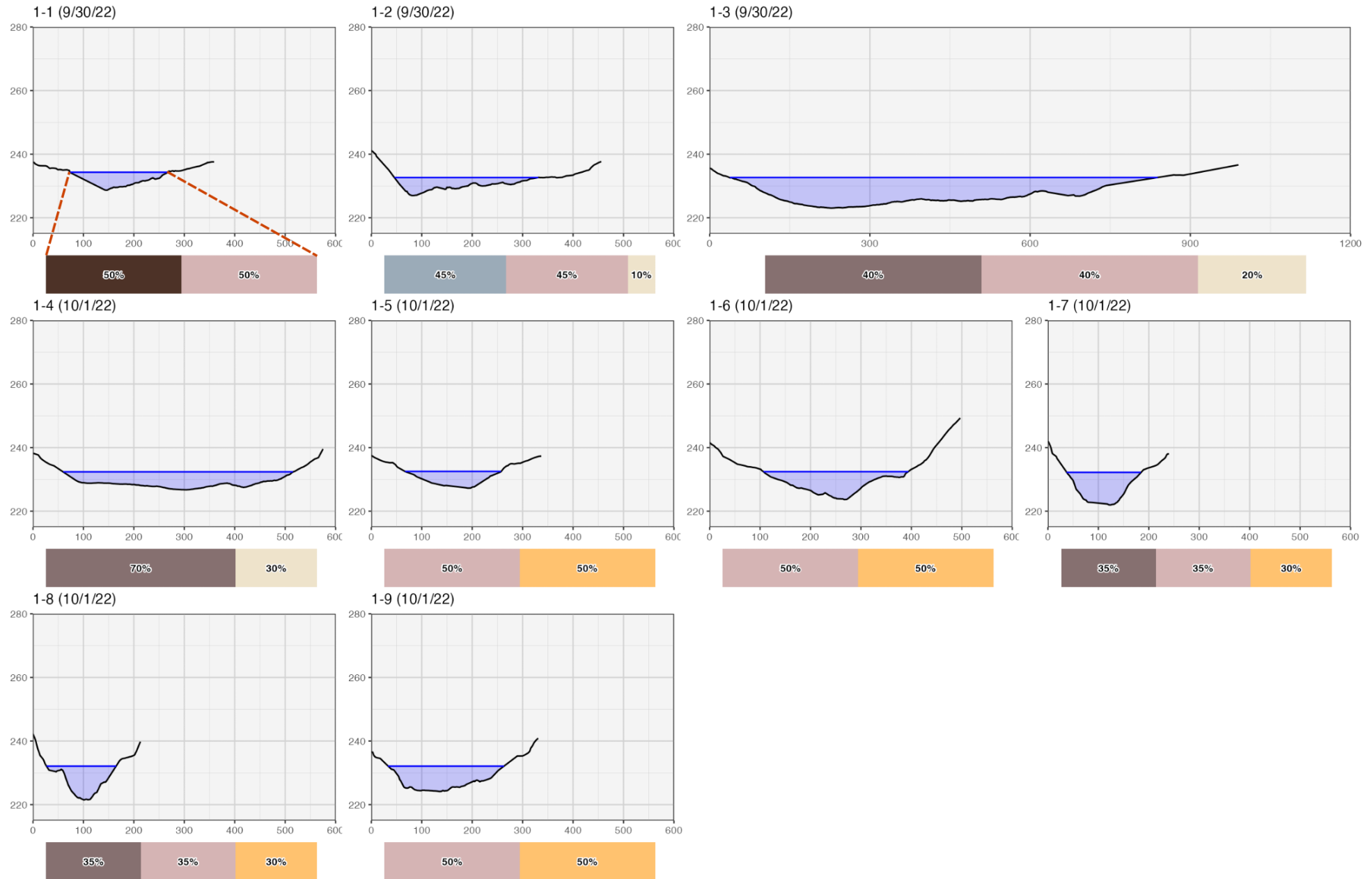
Appendix A – Cross-sections and Substrate Descriptions

Reach 1: Kirkness Rapids to Fidler Lake

Ground Elevation (masl)

Distance from Left Bank (m)

Sediment Type

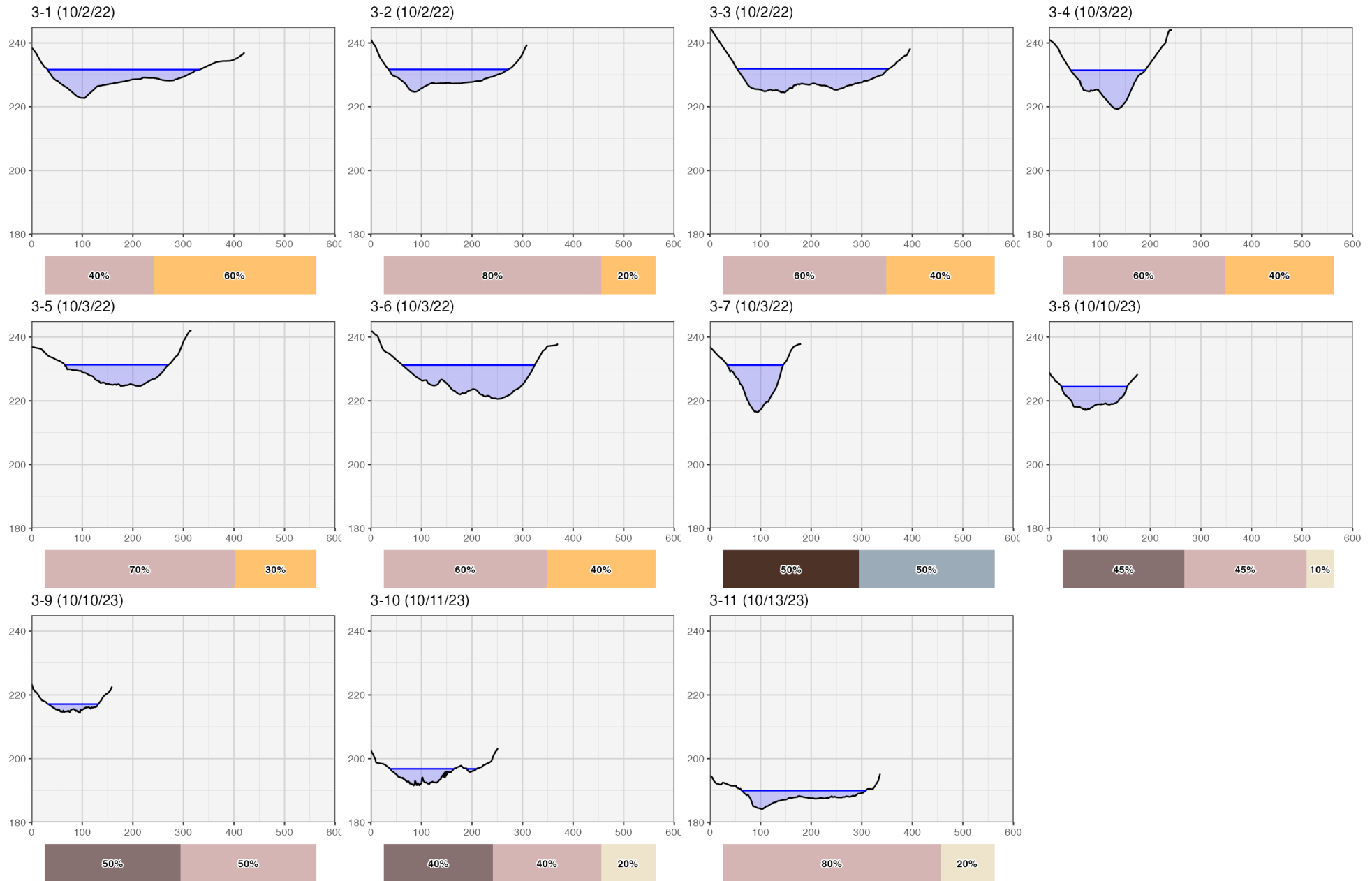


Reach 3: Fidler Lake to Billard Lake

Ground Elevation (masl)

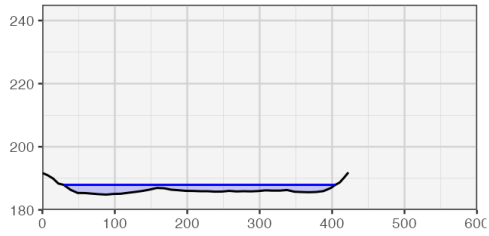
Distance from Left Bank (m)

Sediment Type

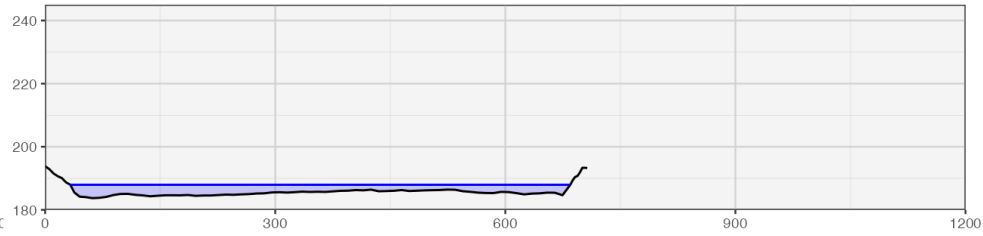


Reach 4: Billard Lake

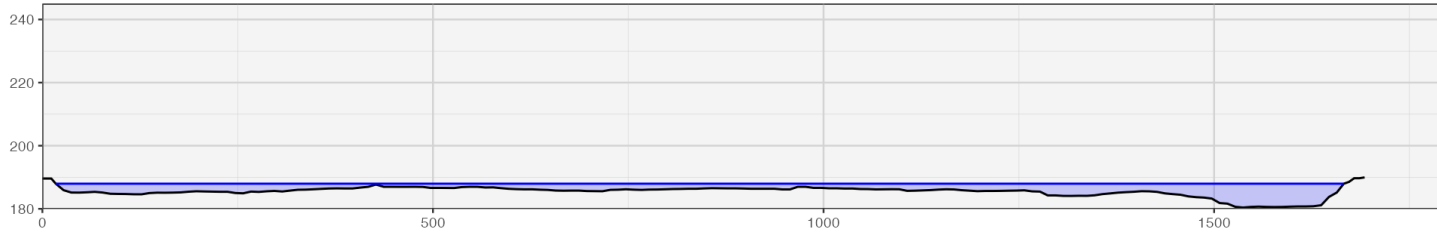
4-1 (7/6/19)



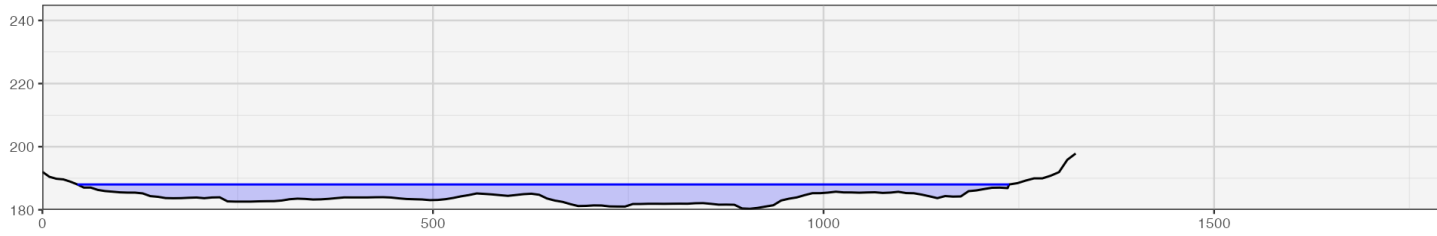
4-2 (7/7/19)



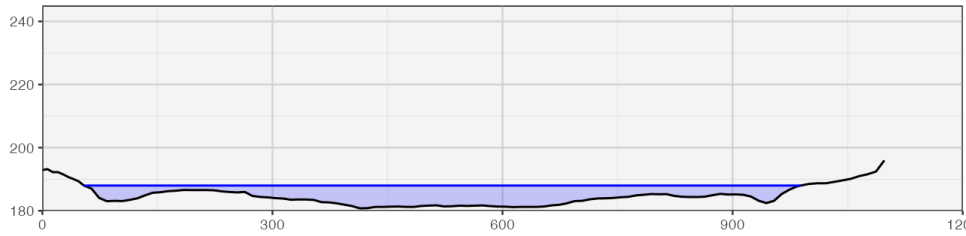
4-3 (7/7/19)



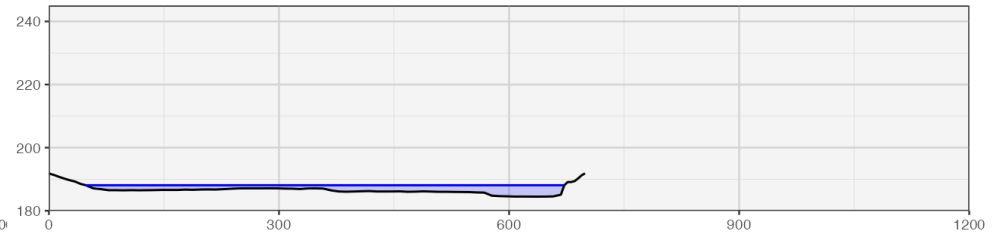
4-4 (7/8/19)



4-5 (7/8/19)



4-6 (7/12/19)



Sediment Type



Bedrock



Boulder



Boulder/Cobble



Cobble



Gravel

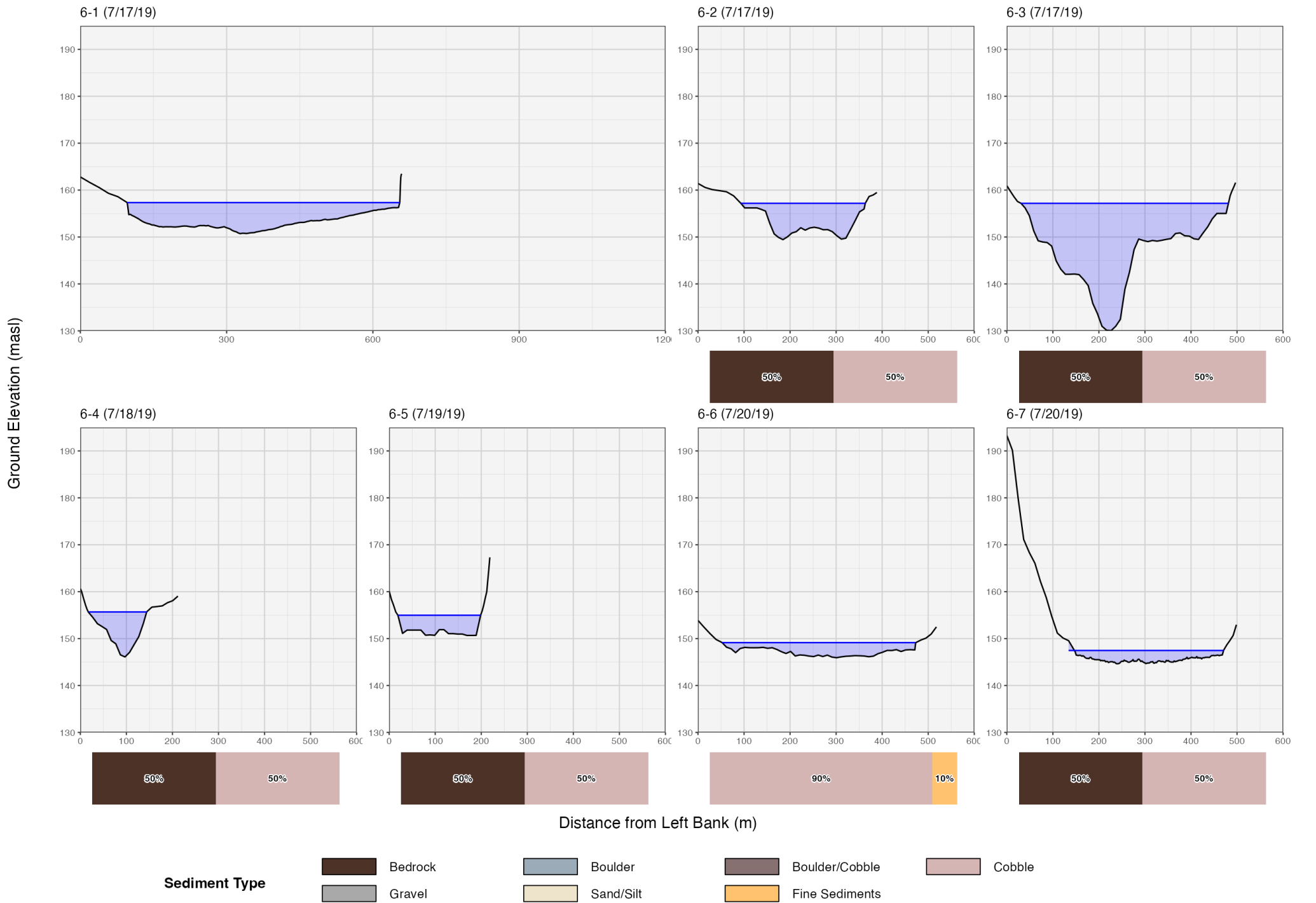


Sand/Silt

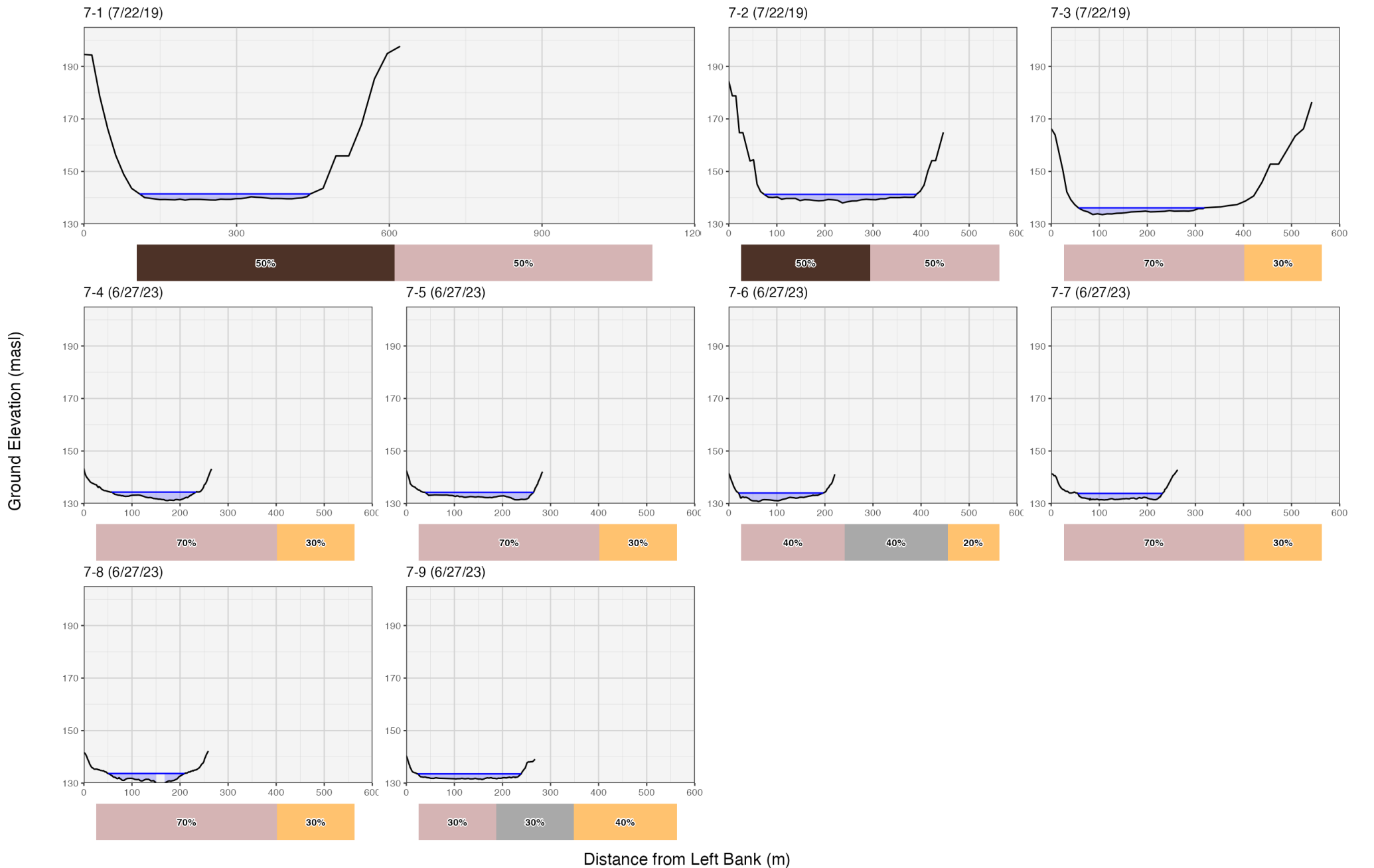


Fine Sediments

Reach 6: Mountain Rapids to The Fours Rapids



Reach 7: The Fours Rapids to the LCR Confluence

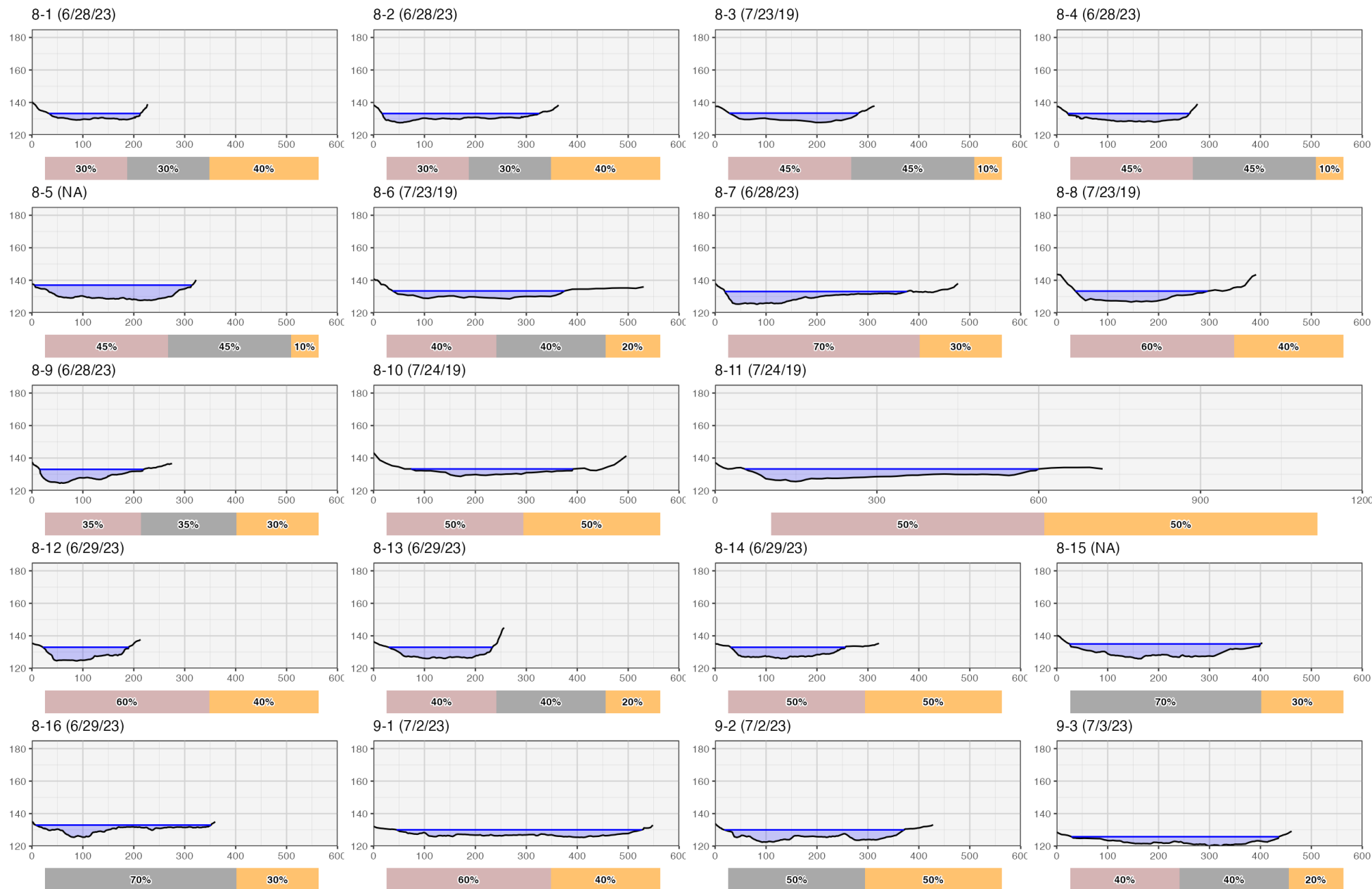


Sediment Type



Reach 8 & 9: The LCR Confluence to Swallow Rapids & Swallow Rapids to Little Beaver Creek

Ground Elevation (masl)



Distance from Left Bank (m)

Sediment Type

